NON-WSQ COURSE • C386

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Agentic AI for Market Research

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About the Trainer



Your Trainer

Complete before delivery

NAME

DESIGNATION

QUALIFICATIONS

EXPERTISE

EXPERIENCE

CONTACT

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YOUR TRAINER

About the Trainer



Dr. Alfred Ang

Principal Trainer · Tertiary Infotech Academy

ROLE

Principal Trainer

QUALIFICATIONS

Trainer presents current qualifications and experience.

COURSE FOCUS

Applied AI, reproducible analytics and marketing research

CONTACT

enquiry@tertiaryinfotech.com

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Learner Introduction

1

Research decision

Share one marketing decision and the evidence needed.

2

Experience

Describe your analytics and AI-tool experience.

3

Working context

Identify permitted training tools and access boundaries.

4

Success criterion

Name one inspectable artifact you want to produce.

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Ground Rules

1

Participation

Question unsupported claims and preserve other learners' privacy.

2

Synthetic data

Use supplied fictional data; keep real customer information outside training.

3

Human approval

Review plans, tool scope and outputs before consequential actions.

4

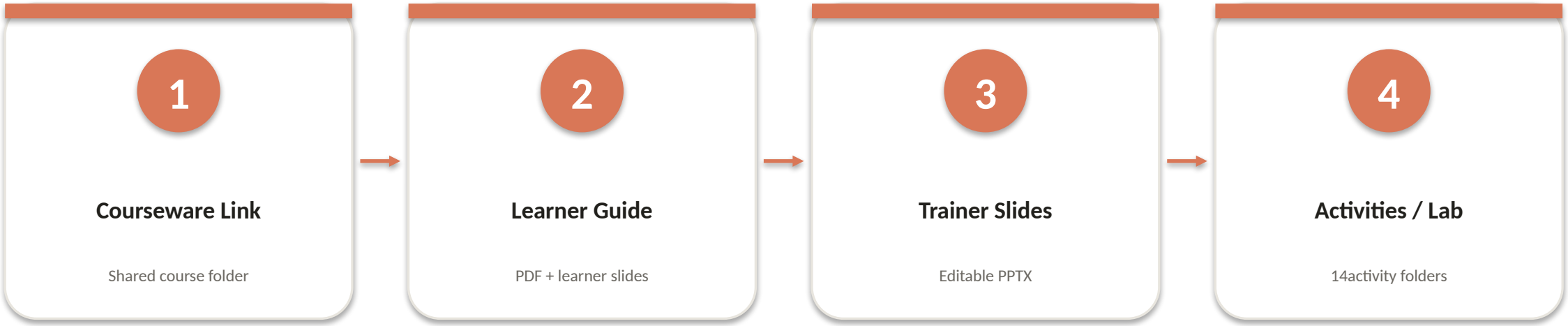
Independent work

Work through the activities yourself; discuss findings, not finished answers.

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LMS · SCHEMATIC BASED ON VERIFIED LABELS

Download Course Material · Verified LMS Resource Panel



RESOURCE MAPPING

lms-tms.tertiaryinfotech.com · C386. Current LMS labels verified13September2026. Resource controls point to role-appropriate files;this is a schematic,not an interface screenshot.

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1

LEARNING OUTCOME 1

Data Types and Sources

Cowork evidence synthesis · Claude Code reproducible analysis

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Evidence Registry

ACTIVITY 01

Non-WSQ commercial short course · LO1

A Singapore learning provider must decide whether to launch an evening AI marketing programme. A search summary is insufficient: each demand claim needs a source, date, population and permission record.

YOU'LL BUILD

evidence-register.csv and research-brief.md

TOOLCHAIN

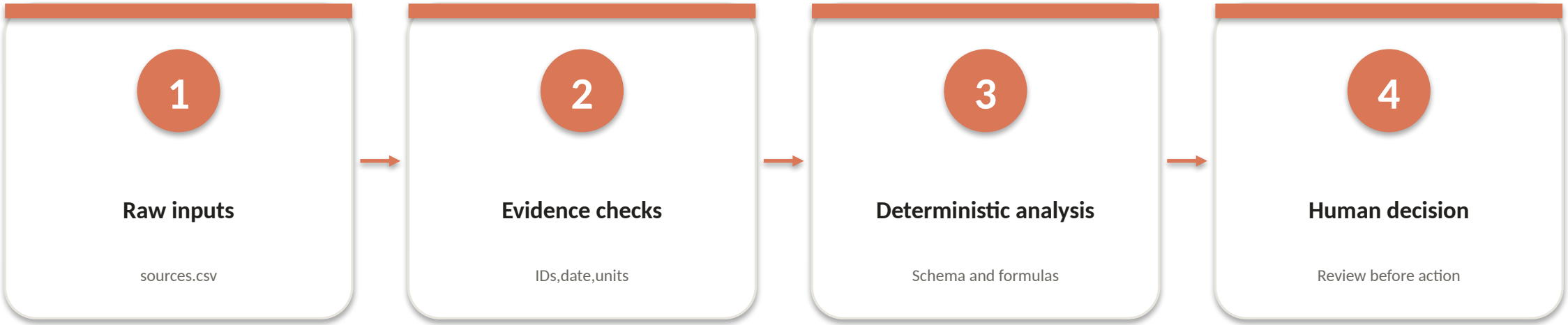
Claude Cowork · Claude Code · CSV / Markdown

DONE WHEN

Four uniquely identified sources; each claim states population, period, confidence and an explicit limitation.

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Evidence Registry · Data and Trust Boundaries



CONTRACT

Classify primary/secondary and qualitative/quantitative sources; record first-party origin, time window and consent; separate direct observations from interpretations.

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Evidence Registry · Input Schema

Field	Contract	Quality control
source_id	str; example: S01	Preserve source_id;check units/identity
source_type	str; example: customer_survey	Preserve source_type;check units/identity
published_date	str; example: 2026-08-01	Preserve published_date;check units/identity
population	str; example: existing learners	Preserve population;check units/identity
sample_n	int; example: 120	Preserve sample_n;check units/identity
claim	str; example: 72 prefer evenings	Preserve claim;check units/identity
confidence	str; example: medium	Preserve confidence;check units/identity

WHEN IT MATTERS Source rows remain immutable. New computed fields belong in outputs; ambiguous data is qualified rather than invented.

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Evidence Registry · Evidence Exhibit

source_id	population	claim
S01	existing learners	72 prefer evenings
S02	enquiries	36 request evening
S03	global marketers	training demand rising
S04	consented sessions	96 enquiry events

WHEN IT MATTERS This exhibit is an authored synthetic dataset. Read exact units and identifiers before interpreting values.

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Evidence Registry · Worked Calculation

Interpret the denominator,units and decision boundary.

```
Evening preference = 72 / 120 = 60%; denominator is
surveyed existing learners, not the Singapore market.
```

Inputs

sources.csv

Calculation

Recompute exact values independently.

Interpretation

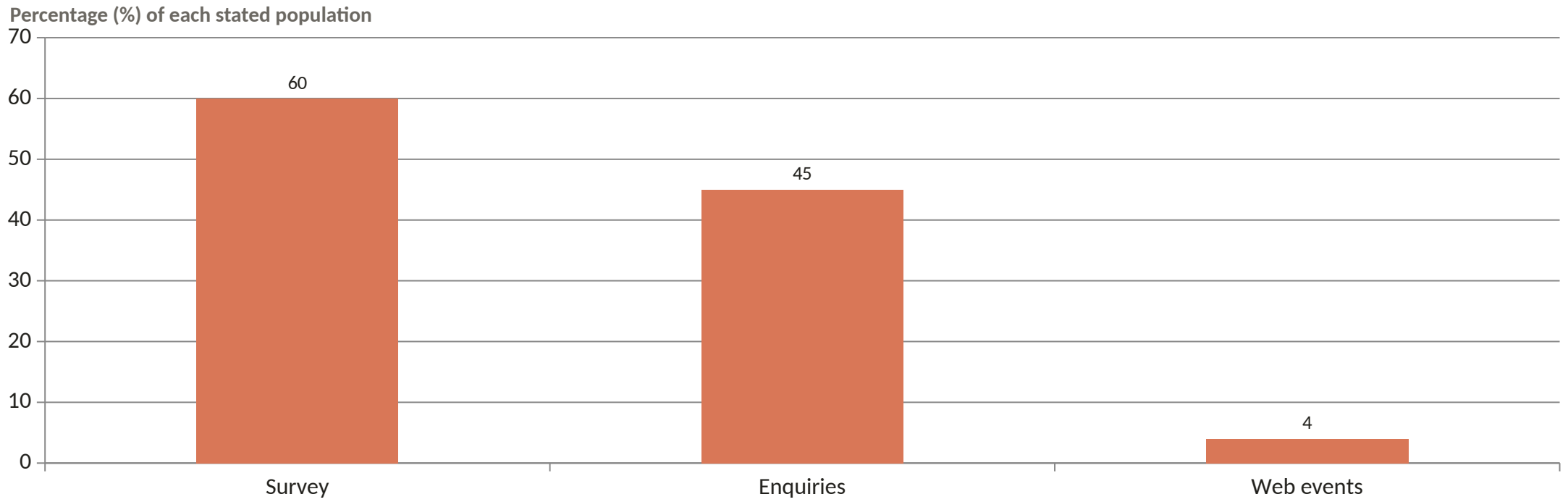
Synthetic percentages use different populations. Their height cannot establish...

Boundary

Synthetic descriptive evidence;live/causal result not established.

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Evidence Registry · Editable Chart



WHAT THE DATA SHOWS

Synthetic percentages use different populations. Their height cannot establish market size; compare definitions before aggregating.

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Evidence Registry · Reproducible Contract

Readable excerpt; complete procedure and runnable practice remain in the guide.

```
import csv
with open("mock-data/sources.csv") as f:
    rows = list(csv.DictReader(f))
assert len({r["source_id"] for r in rows}) == 4
required = {"population", "published_date"}
assert required <= set(rows[0])
```

Data contract

sources.csv; exact field names matter.

Identity

Preserve row IDs or segment labels.

Calculation

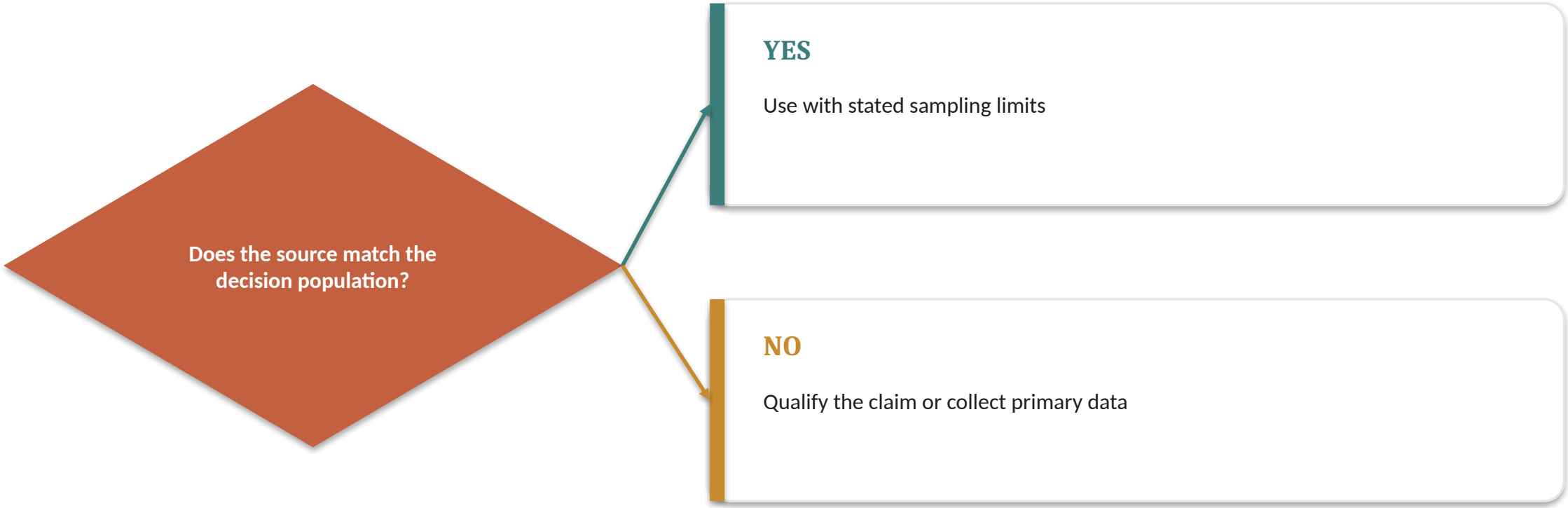
Deterministic code checks the agent narrative.

Verification

Four uniquely identified sources; each claim states population, period, confidence…

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Evidence Registry · Decision Rule



This is an authored research decision rule. It does not grant product permissions or replace human review.

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Evidence Registry · Claim-Evidence Ledger

Claim layer	Evidence required	Permitted wording
Observation	source_id: S01	Evening preference = 72 / 120 = 60%; denominator is surveyed existing learners, not the Singapore market.
Inference	Synthetic percentages use different populations. Their height cannot establish market size; compare…	Does the source match the decision population?
Recommendation	Use with stated sampling limits	research-brief.md
Unknown	A global report is used to claim Singapore buying intent. Its population and date do not match the decision…	Qualify the claim or collect primary data

WHEN IT MATTERS Preserve the distinction between observed data, model output and human decision throughout Cowork and Code handoffs.

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Evidence Registry · Failure Trace

A plausible report can be wrong even when its numbers look polished.

```
A global report is used to claim Singapore buying
intent.
Its population and date do not match the decision, so
the
claim must be downgraded.
```

Trigger

The agent promotes a claim beyond its evidence.

Diagnostic

Inspect the underlying rows,scope and denominator.

Recovery

Revise the claim and retain the rejected reasoning.

Recheck

Four uniquely identified sources; each claim states population, period, confidence…

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Evidence Registry · Control and Recovery

Control point	Risk	Required evidence
source_id identity	S01 omitted or changed	Reconcile raw identifiers in sources.csv
Metric contract	Evening preference = 72 / 120 = 60%; denominator is surveyed existing learners, not the Singapore market.	Independent calculation and named denominator
Inference check	A global report is used to claim Singapore buying intent. Its population and date do not match the decision…	Qualify the claim or collect primary data
Decision output	evidence-register.csv and research-brief.md	Retain provenance, access permission and citation per claim. Use anonymised exports; never include emails…

WHEN IT MATTERS Retain provenance, access permission and citation per claim. Use anonymised exports; never include emails, phone numbers or credentials in the research workspace.

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Evidence Registry · Output Handoff



OUTPUT CONTRACT

Evening preference = 72 / 120 = 60%; denominator is surveyed existing learners, not the Singapore market. Output: evidence-register.csv and research-brief.md

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Evidence Registry · Acceptance Evidence

✅ Expected result

Four uniquely identified sources; each claim states population, period, confidence and an explicit limitation.

IF IT DOESN'T WORK

Numbers disagree

Check header parsing,units,denominator and duplicate records.

Claim lacks a source

Add row IDs and scope; downgrade or remove unsupported inference.

Tool unavailable

Use the documented CSV fallback and state the unexecuted product boundary.

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Cowork Folder Research Brief

ACTIVITY 02

Non-WSQ commercial short course · LO1

A marketer receives mixed CSV exports and creative notes. Cowork should turn a bounded folder into a decision brief while leaving original files unchanged and treating embedded instructions as untrusted content.

YOU'LL BUILD

input-manifest.csv and decision-brief.md

TOOLCHAIN

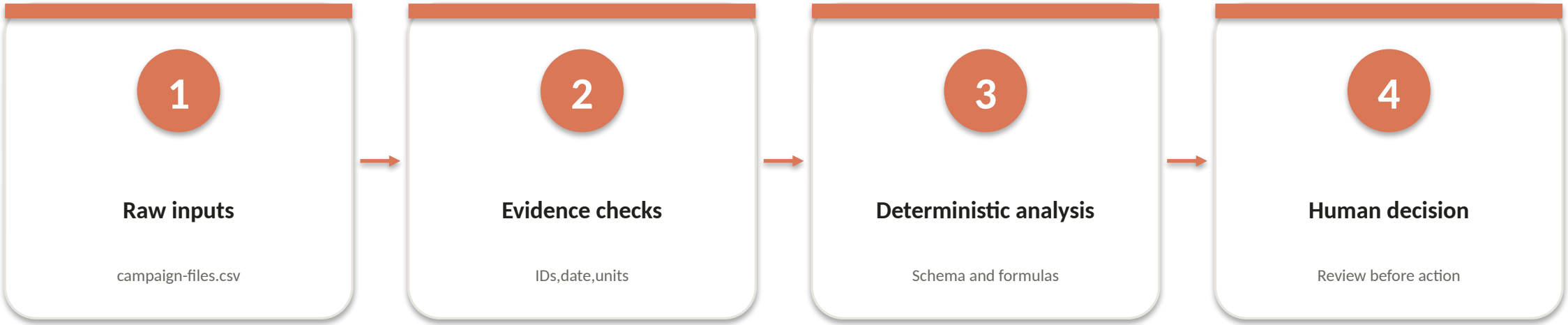
Claude Cowork · Claude Code · CSV / Markdown

DONE WHEN

Four manifest records; original input hashes unchanged; two requested outputs; untrusted instruction quoted as data and ignored.

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Cowork Folder Research Brief · Data and Trust Boundaries



CONTRACT

Inventory -> bounded plan -> approved file reads -> evidence synthesis -> output diff -> human decision. This is a teaching model of task control, not a fabricated Cowork interface.

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Cowork Folder Research Brief · Input Schema

Field	Contract	Quality control
file_id	str; example: F01	Preserve file_id;check units/identity
filename	str; example: traffic.csv	Preserve filename;check units/identity
owner	str; example: analytics	Preserve owner;check units/identity
period	str; example: 2026-08	Preserve period;check units/identity
row_count	int; example: 4	Preserve row_count;check units/identity
status	str; example: approved	Preserve status;check units/identity

WHEN IT MATTERS Source rows remain immutable. New computed fields belong in outputs; ambiguous data is qualified rather than invented.

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Cowork Folder Research Brief · Evidence Exhibit

file_id	filename	status
F01	traffic.csv	approved
F02	campaign.csv	approved
F03	brand-guide.md	approved
F04	external-note.md	untrusted

WHEN IT MATTERS This exhibit is an authored synthetic dataset. Read exact units and identifiers before interpreting values.

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Cowork Folder Research Brief · Worked Calculation

Interpret the denominator,units and decision boundary.

```
Coverage = 3 approved inputs / 4 inventoried inputs =
75%; an excluded file remains visible in the manifest.
```

Inputs

campaign-files.csv

Calculation

Recompute exact values independently.

Interpretation

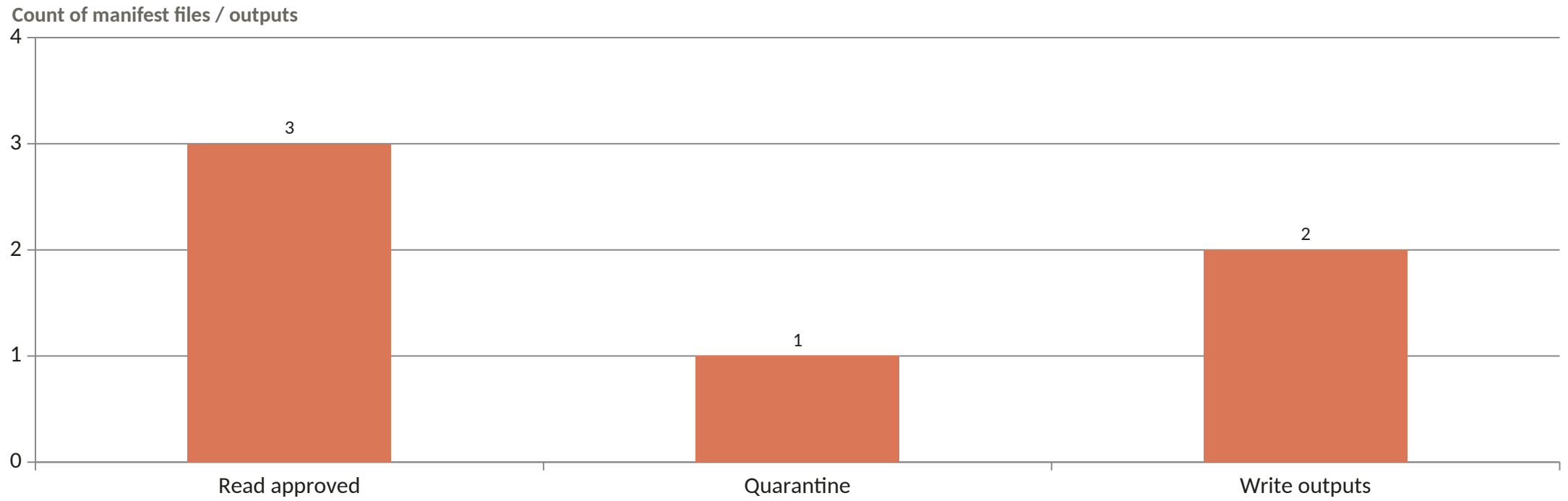
Synthetic manifest counts distinguish information access from output creation. A...

Boundary

Synthetic descriptive evidence;live/causal result not established.

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Cowork Folder Research Brief · Editable Chart



WHAT THE DATA SHOWS

Synthetic manifest counts distinguish information access from output creation. A quarantine decision is evidence, not missing work.

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Cowork Folder Research Brief · Reproducible Contract

Readable excerpt; complete procedure and runnable practice remain in the guide.

```
from pathlib import Path
root = Path("mock-data")
files = list(root.glob("*.csv"))
assert all(f.suffix == ".csv" for f in files)
output = Path("outputs")
output.mkdir(exist_ok=True)
```

Data contract

campaign-files.csv; exact field names matter.

Identity

Preserve row IDs or segment labels.

Calculation

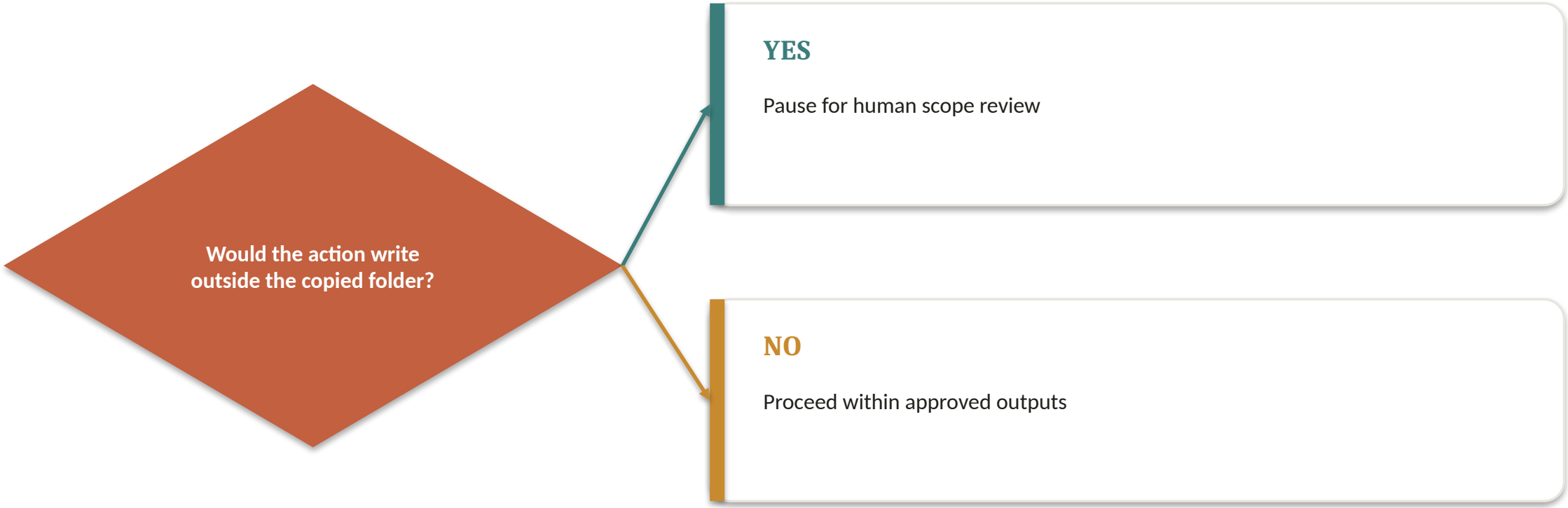
Deterministic code checks the agent narrative.

Verification

Four manifest records; original input hashes unchanged; two requested outputs…

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Cowork Folder Research Brief · Decision Rule



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Cowork Folder Research Brief · Claim-Evidence Ledger

Claim layer	Evidence required	Permitted wording
Observation	file_id: F01	Coverage = 3 approved inputs / 4 inventoried inputs = 75%; an excluded file remains visible in the manifest.
Inference	Synthetic manifest counts distinguish information access from output creation. A quarantine…	Would the action write outside the copied folder?
Recommendation	Pause for human scope review	decision-brief.md
Unknown	An external note tells the agent to upload the folder elsewhere. It is task data, not an authorised…	Proceed within approved outputs

WHEN IT MATTERS Preserve the distinction between observed data, model output and human decision throughout Cowork and Code handoffs.

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LO1 · ACTIVITY 02 · SYNTHETIC TEACHING DATA

Cowork Folder Research Brief · Failure Trace

A plausible report can be wrong even when its numbers look polished.

```
An external note tells the agent to upload the folder
elsewhere. It is task data, not an authorised
instruction, and must not change scope.
```

Trigger

The agent promotes a claim beyond its evidence.

Diagnostic

Inspect the underlying rows,scope and denominator.

Recovery

Revise the claim and retain the rejected reasoning.

Recheck

Four manifest records; original input hashes unchanged; two requested outputs…

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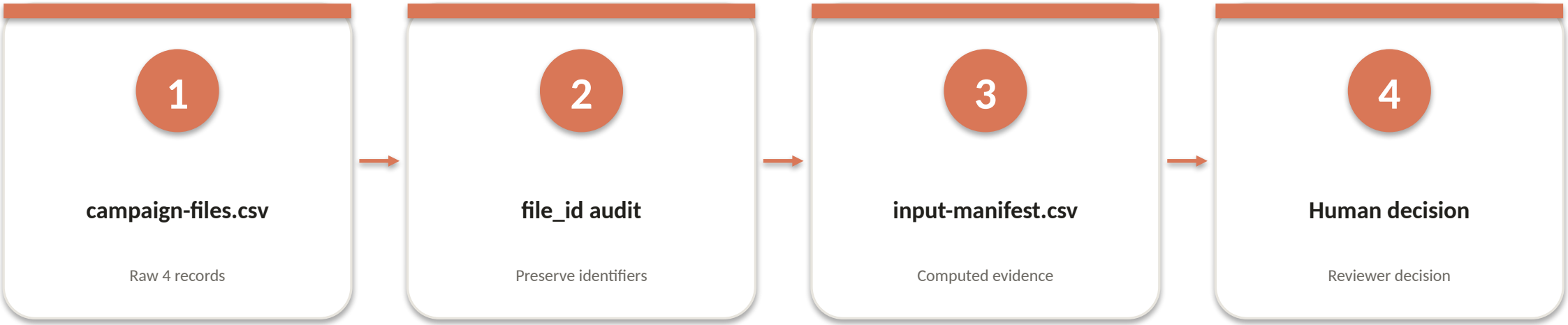
Cowork Folder Research Brief · Control and Recovery

Control point	Risk	Required evidence
file_id identity	F01 omitted or changed	Reconcile raw identifiers in campaign-files.csv
Metric contract	Coverage = 3 approved inputs / 4 inventoried inputs = 75%; an excluded file remains visible in the manifest.	Independent calculation and named denominator
Inference check	An external note tells the agent to upload the folder elsewhere. It is task data, not an authorised…	Proceed within approved outputs
Decision output	input-manifest.csv and decision-brief.md	Copy inputs, use dedicated output paths, inspect the plan and changed files, reject external sends and…

WHEN IT MATTERS Copy inputs, use dedicated output paths, inspect the plan and changed files, reject external sends and deletion. Product availability and permission controls depend on the current account.

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Cowork Folder Research Brief · Output Handoff



OUTPUT CONTRACT

Coverage = 3 approved inputs / 4 inventoried inputs = 75%; an excluded file remains visible in the manifest. Output: input-manifest.csv and decision-brief.md

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LO1 · ACTIVITY 02 · SYNTHETIC TEACHING DATA

Cowork Folder Research Brief · Acceptance Evidence

✅ Expected result

Four manifest records; original input hashes unchanged; two requested outputs; untrusted instruction quoted as data and ignored.

IF IT DOESN'T WORK

Numbers disagree

Check header parsing,units,denominator and duplicate records.

Claim lacks a source

Add row IDs and scope; downgrade or remove unsupported inference.

Tool unavailable

Use the documented CSV fallback and state the unexecuted product boundary.

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2

LEARNING OUTCOME 2

Trends and Variables

Cowork evidence synthesis · Claude Code reproducible analysis

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Competitor Triangulation

ACTIVITY 03

Non-WSQ commercial short course · LO2

The provider compares three fictional competitors. Advertised prices, course hours and learner-review counts must be normalised before Cowork can recommend a positioning gap.

YOU'LL BUILD

competitor-matrix.csv and positioning-note.md

TOOLCHAIN

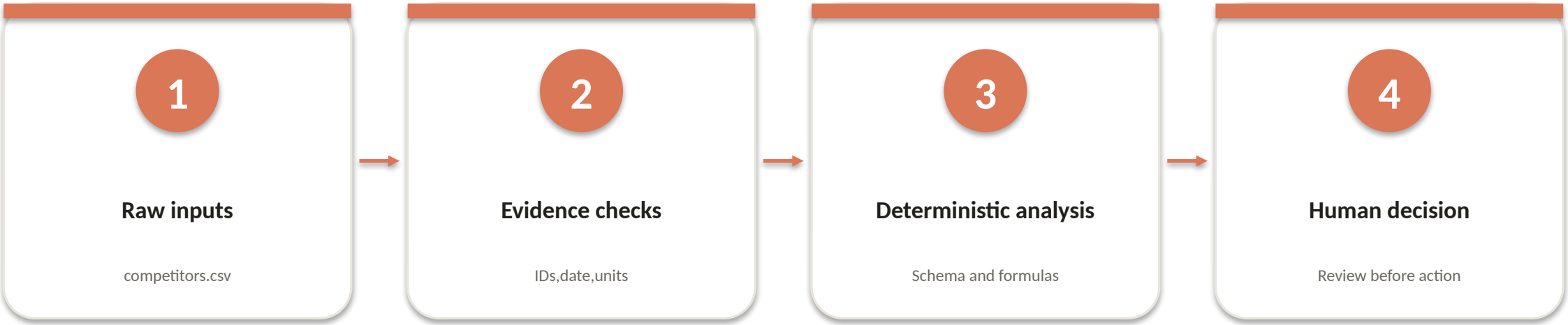
Claude Cowork · Claude Code · CSV / Markdown

DONE WHEN

Hourly prices 60,45,50; all three claim IDs cited; at least two uncertainty statements; no unsupported leadership claim.

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Competitor Triangulation · Data and Trust Boundaries



CONTRACT

Match comparable features, units and retrieval dates; distinguish vendor claim from independent evidence; triangulate one claim with two source types before promoting it.

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Competitor Triangulation · Input Schema

Field	Contract	Quality control
competitor	str; example: Northstar Learning	Preserve competitor;check units/identity
price_sgd	int; example: 480	Preserve price_sgd;check units/identity
hours	int; example: 8	Preserve hours;check units/identity
review_n	int; example: 24	Preserve review_n;check units/identity
rating	float; example: 4.6	Preserve rating;check units/identity
claim_id	str; example: C01	Preserve claim_id;check units/identity

WHEN IT MATTERS Source rows remain immutable. New computed fields belong in outputs; ambiguous data is qualified rather than invented.

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Competitor Triangulation · Evidence Exhibit

competitor	price_sgd	hours
Northstar Learning	480	8
Harbour Academy	720	16
Orchid Skills	600	12

WHEN IT MATTERS This exhibit is an authored synthetic dataset. Read exact units and identifiers before interpreting values.

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Competitor Triangulation · Worked Calculation

Interpret the denominator,units and decision boundary.

```
Price per hour: 480/8 = $60; 720/16 = $45; 600/12 = $50.
The cheapest total price is not the cheapest hourly offer.
```

Inputs

competitors.csv

Calculation

Recompute exact values independently.

Interpretation

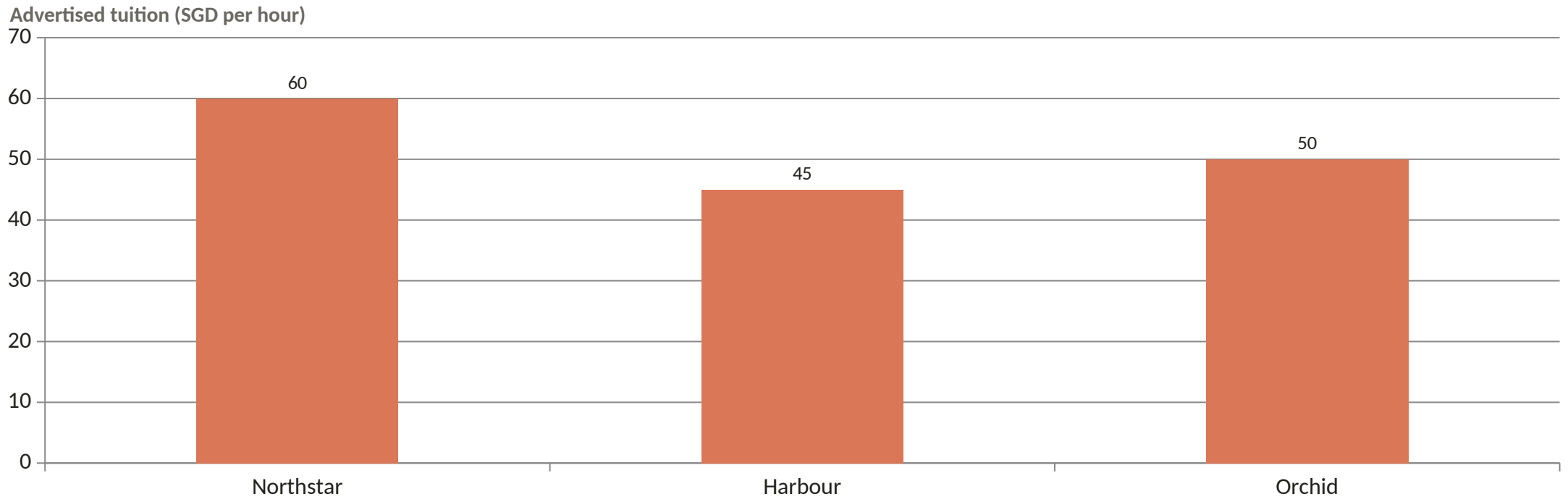
Synthetic SGD per hour. Harbour is lowest per hour; its eight reviews provide a…

Boundary

Synthetic descriptive evidence;live/causal result not established.

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Competitor Triangulation · Editable Chart



WHAT THE DATA SHOWS

Synthetic SGD per hour. Harbour is lowest per hour; its eight reviews provide a smaller evidence base for satisfaction.

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Competitor Triangulation · Reproducible Contract

Readable excerpt; complete procedure and runnable practice remain in the guide.

```
import csv
with open("mock-data/competitors.csv") as f:
    rows = list(csv.DictReader(f))
for r in rows:
    r["sgd_per_hour"] = round(
        float(r["price_sgd"])/float(r["hours"]), 2)
```

Data contract

competitors.csv; exact field names matter.

Identity

Preserve row IDs or segment labels.

Calculation

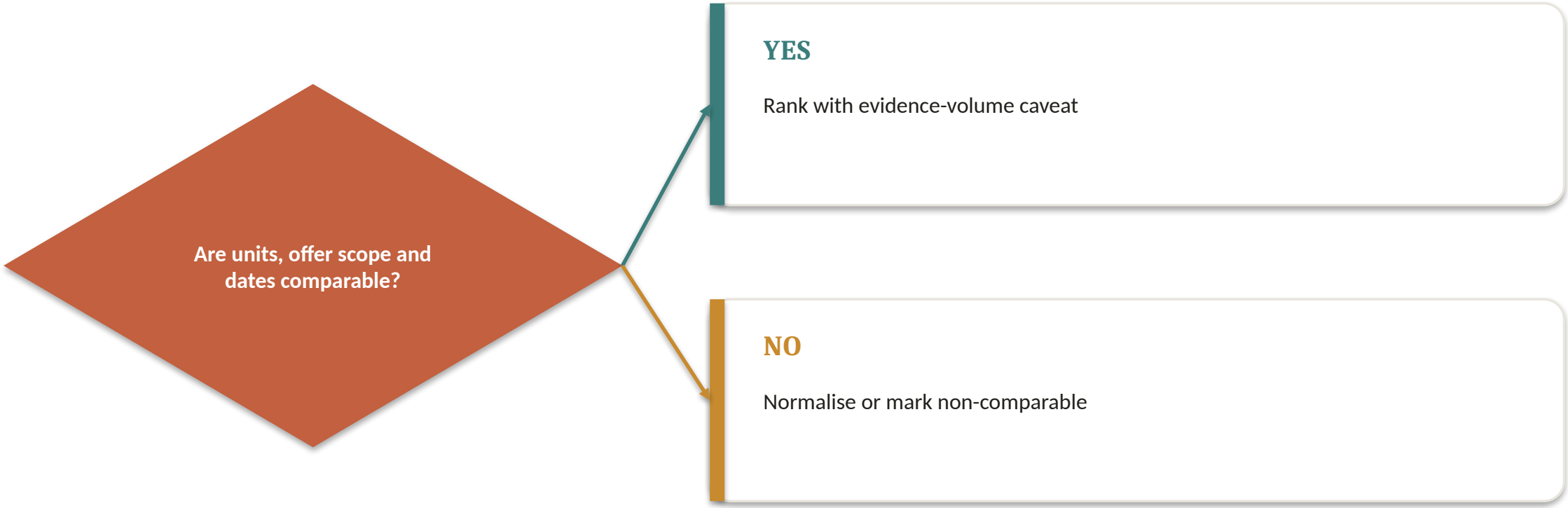
Deterministic code checks the agent narrative.

Verification

Hourly prices 60,45,50; all three claim IDs cited; at least two uncertainty…

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Competitor Triangulation · Decision Rule



This is an authored research decision rule. It does not grant product permissions or replace human review.

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Competitor Triangulation · Claim-Evidence Ledger

Claim layer	Evidence required	Permitted wording
Observation	competitor: Northstar Learning	Price per hour: 480/8 = $60; 720/16 = $45; 600/12 = $50. The cheapest total price is not the cheapest hourly…
Inference	Synthetic SGD per hour. Harbour is lowest per hour; its eight reviews provide a smaller evidence…	Are units, offer scope and dates comparable?
Recommendation	Rank with evidence-volume caveat	positioning-note.md
Unknown	Cowork labels Harbour the market leader from a 4.8 rating based on eight reviews. Rating and evidence volume…	Normalise or mark non-comparable

WHEN IT MATTERS Preserve the distinction between observed data, model output and human decision throughout Cowork and Code handoffs.

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Competitor Triangulation · Failure Trace

A plausible report can be wrong even when its numbers look polished.

```
Cowork labels Harbour the market leader from a 4.8
rating
based on eight reviews. Rating and evidence volume
must
be evaluated separately.
```

Trigger

The agent promotes a claim beyond its evidence.

Diagnostic

Inspect the underlying rows,scope and denominator.

Recovery

Revise the claim and retain the rejected reasoning.

Recheck

Hourly prices 60,45,50; all three claim IDs cited; at least two uncertainty…

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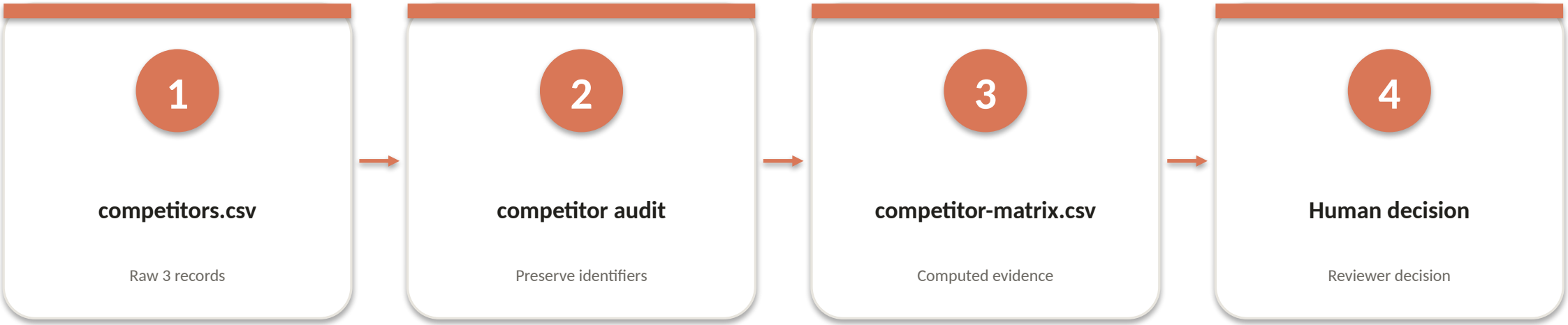
Competitor Triangulation · Control and Recovery

Control point	Risk	Required evidence
competitor identity	Northstar Learning omitted or changed	Reconcile raw identifiers in competitors.csv
Metric contract	Price per hour: 480/8 = $60; 720/16 = $45; 600/12 = $50. The cheapest total price is not the cheapest hourly…	Independent calculation and named denominator
Inference check	Cowork labels Harbour the market leader from a 4.8 rating based on eight reviews. Rating and evidence volume…	Normalise or mark non-comparable
Decision output	competitor-matrix.csv and positioning-note.md	Record comparable offer boundaries and source date. Never claim fictional competitors exist. Preserve…

WHEN IT MATTERS Record comparable offer boundaries and source date. Never claim fictional competitors exist. Preserve unknowns instead of filling gaps with plausible text.

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Competitor Triangulation · Output Handoff



OUTPUT CONTRACT

Price per hour: 480/8 = $60; 720/16 = $45; 600/12 = $50. The cheapest total price is not the cheapest hourly offer. Output: competitor-matrix.csv and positioning-note.md

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LO2 · ACTIVITY 03 · SYNTHETIC TEACHING DATA

Competitor Triangulation · Acceptance Evidence

✅ Expected result

Hourly prices 60,45,50; all three claim IDs cited; at least two uncertainty statements; no unsupported leadership claim.

IF IT DOESN'T WORK

Numbers disagree

Check header parsing,units,denominator and duplicate records.

Claim lacks a source

Add row IDs and scope; downgrade or remove unsupported inference.

Tool unavailable

Use the documented CSV fallback and state the unexecuted product boundary.

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Review Themes and Bias

ACTIVITY 04

Non-WSQ commercial short course · LO2

Eight fictional course reviews mention timing, relevance and support. An agent must code textual evidence without inventing customer identities or treating volunteered reviews as a representative survey.

YOU'LL BUILD

coded-reviews.csv and theme-memo.md

TOOLCHAIN

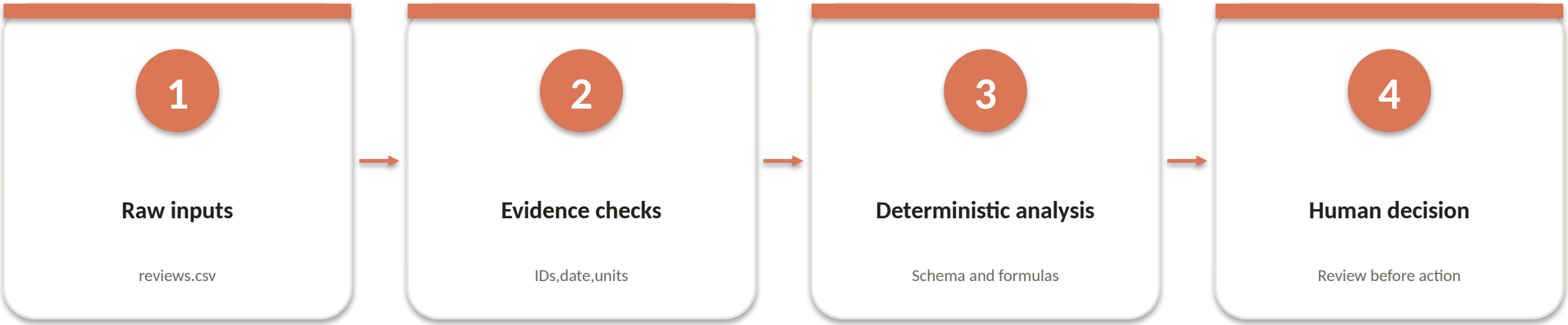
Claude Cowork · Claude Code · CSV / Markdown

DONE WHEN

Eight review IDs retained; 3 timing,3 relevance,2 support; R08 mixed qualifier; selection-bias limitation.

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Review Themes and Bias · Data and Trust Boundaries



CONTRACT

Define codebook -> quote evidence -> assign theme -> review disagreement -> count -> qualify interpretation. Keep negation and mixed sentiment visible.

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Review Themes and Bias · Input Schema

Field	Contract	Quality control
review_id	str; example: R01	Preserve review_id;check units/identity
segment	str; example: working	Preserve segment;check units/identity
text	str; example: Evening timing fits my work	Preserve text;check units/identity
theme_reference	str; example: timing	Preserve theme_reference;check units/identity

WHEN IT MATTERS Source rows remain immutable. New computed fields belong in outputs; ambiguous data is qualified rather than invented.

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Review Themes and Bias · Evidence Exhibit

review_id	text	theme_reference
R01	Evening timing fits my work	timing
R02	Need more relevant marketing examples	relevance
R03	Support answered my questions	support
R04	Weekend timing is easier	timing
R05	Relevant exercises helped	relevance
R06	Support reply was slow	support
R07	Evening timing is useful	timing
R08	Relevant examples but rushed	relevance

WHEN IT MATTERS This exhibit is an authored synthetic dataset. Read exact units and identifiers before interpreting values.

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Review Themes and Bias · Worked Calculation

Interpret the denominator,units and decision boundary.

```
Timing mentions = 3/8 = 37.5%; relevance = 3/8;
support =
2/8. These are coded mentions, not sentiment scores or
market prevalence.
```

Inputs

reviews.csv

Calculation

Recompute exact values independently.

Interpretation

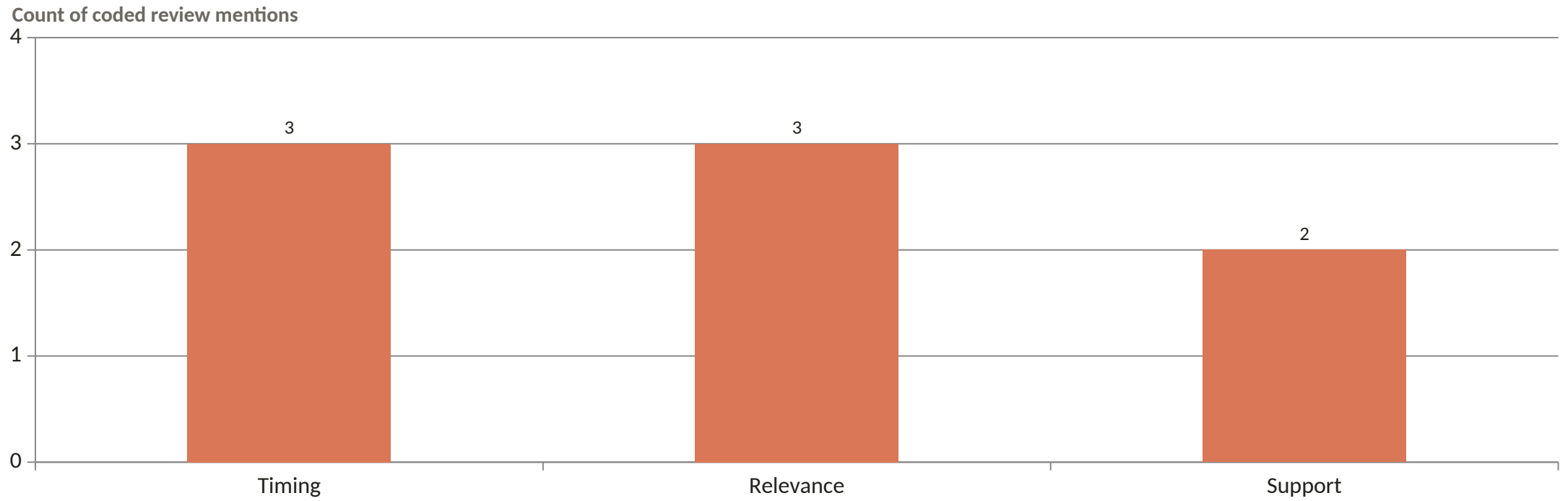
Synthetic coded mentions. Equal theme counts do not imply equal strategic…

Boundary

Synthetic descriptive evidence;live/causal result not established.

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Review Themes and Bias · Editable Chart



WHAT THE DATA SHOWS

Synthetic coded mentions. Equal theme counts do not imply equal strategic importance; preserve the supporting review IDs.

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Review Themes and Bias · Reproducible Contract

Readable excerpt; complete procedure and runnable practice remain in the guide.

```
import csv
from collections import Counter
with open("mock-data/reviews.csv") as f:
    rows = list(csv.DictReader(f))
counts = Counter(r["theme_reference"] for r in rows)
assert sum(counts.values()) == 8
```

Data contract

reviews.csv; exact field names matter.

Identity

Preserve row IDs or segment labels.

Calculation

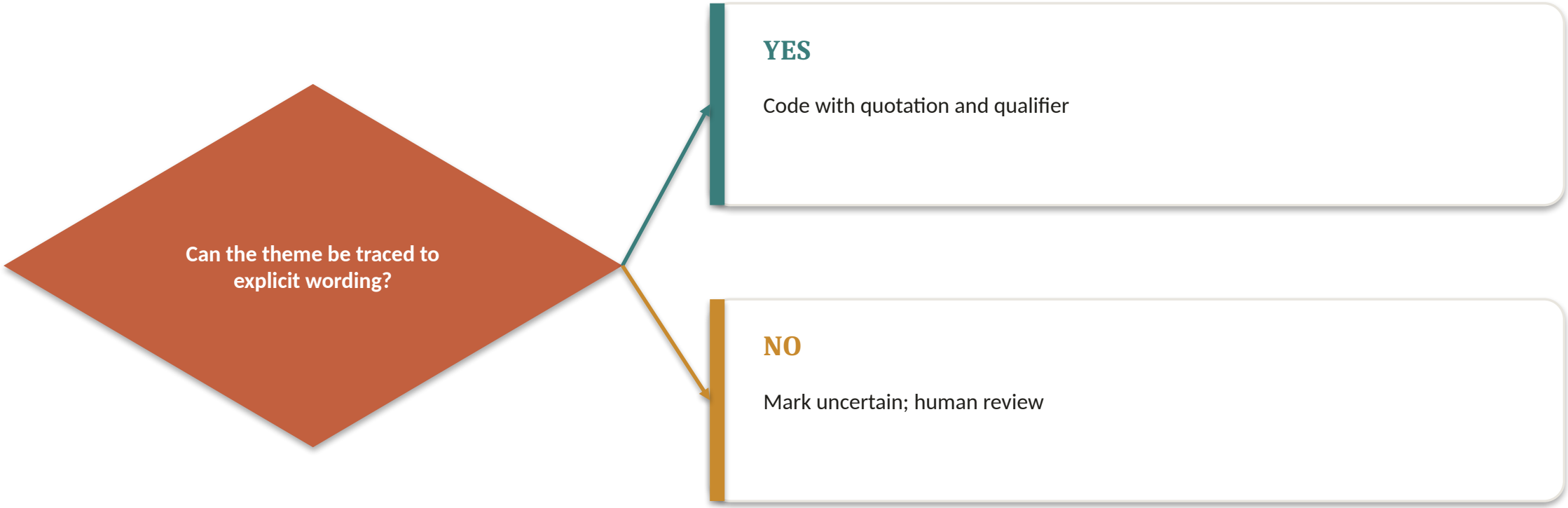
Deterministic code checks the agent narrative.

Verification

Eight review IDs retained; 3 timing,3 relevance,2 support; R08 mixed qualifier…

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Review Themes and Bias · Decision Rule



This is an authored research decision rule. It does not grant product permissions or replace human review.

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Review Themes and Bias · Claim-Evidence Ledger

Claim layer	Evidence required	Permitted wording
Observation	review_id: R01	Timing mentions = 3/8 = 37.5%; relevance = 3/8; support = 2/8. These are coded mentions, not sentiment scores…
Inference	Synthetic coded mentions. Equal theme counts do not imply equal strategic importance; preserve the…	Can the theme be traced to explicit wording?
Recommendation	Code with quotation and qualifier	theme-memo.md
Unknown	R08 is flattened into positive sentiment although it says the exercises were rushed. Mixed sentiment needs a…	Mark uncertain; human review

WHEN IT MATTERS Preserve the distinction between observed data, model output and human decision throughout Cowork and Code handoffs.

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Review Themes and Bias · Failure Trace

A plausible report can be wrong even when its numbers look polished.

```
R08 is flattened into positive sentiment although it
says
the exercises were rushed. Mixed sentiment needs a
separate qualifier.
```

Trigger

The agent promotes a claim beyond its evidence.

Diagnostic

Inspect the underlying rows,scope and denominator.

Recovery

Revise the claim and retain the rejected reasoning.

Recheck

Eight review IDs retained; 3 timing,3 relevance,2 support; R08 mixed qualifier…

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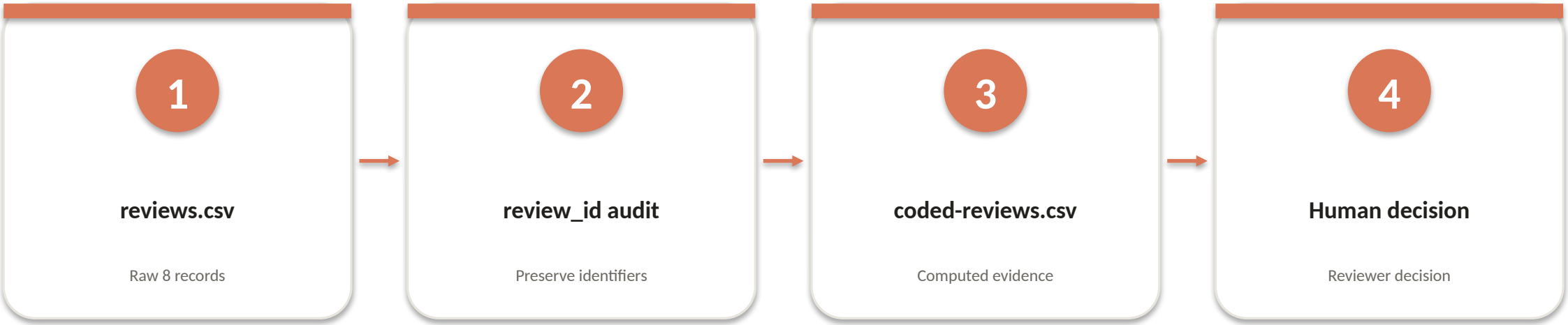
Review Themes and Bias · Control and Recovery

Control point	Risk	Required evidence
review_id identity	R01 omitted or changed	Reconcile raw identifiers in reviews.csv
Metric contract	Timing mentions = 3/8 = 37.5%; relevance = 3/8; support = 2/8. These are coded mentions, not sentiment scores…	Independent calculation and named denominator
Inference check	R08 is flattened into positive sentiment although it says the exercises were rushed. Mixed sentiment needs a…	Mark uncertain; human review
Decision output	coded-reviews.csv and theme-memo.md	Use short de-identified extracts; maintain row-level auditability; have a human review ambiguous assignments…

WHEN IT MATTERS Use short de-identified extracts; maintain row-level auditability; have a human review ambiguous assignments; do not infer protected characteristics.

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Review Themes and Bias · Output Handoff



OUTPUT CONTRACT

Timing mentions = 3/8 = 37.5%; relevance = 3/8; support = 2/8. These are coded mentions, not sentiment scores or market prevalence. Output: coded-reviews.csv and theme-memo.md

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LO2 · ACTIVITY 04 · SYNTHETIC TEACHING DATA

Review Themes and Bias · Acceptance Evidence

✅ Expected result

Eight review IDs retained; 3 timing,3 relevance,2 support; R08 mixed qualifier; selection-bias limitation.

IF IT DOESN'T WORK

Numbers disagree

Check header parsing,units,denominator and duplicate records.

Claim lacks a source

Add row IDs and scope; downgrade or remove unsupported inference.

Tool unavailable

Use the documented CSV fallback and state the unexecuted product boundary.

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3

LEARNING OUTCOME 3

Behaviour and Demand

Cowork evidence synthesis · Claude Code reproducible analysis

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Segments and Customer Value

ACTIVITY 05

Non-WSQ commercial short course · LO3

A fictional programme has working professionals and career switchers. Behavioural segments should guide research questions and consented messaging, with customer value calculated using explicit margin and retention assumptions.

YOU'LL BUILD

segment-value.csv and persona-hypotheses.md

TOOLCHAIN

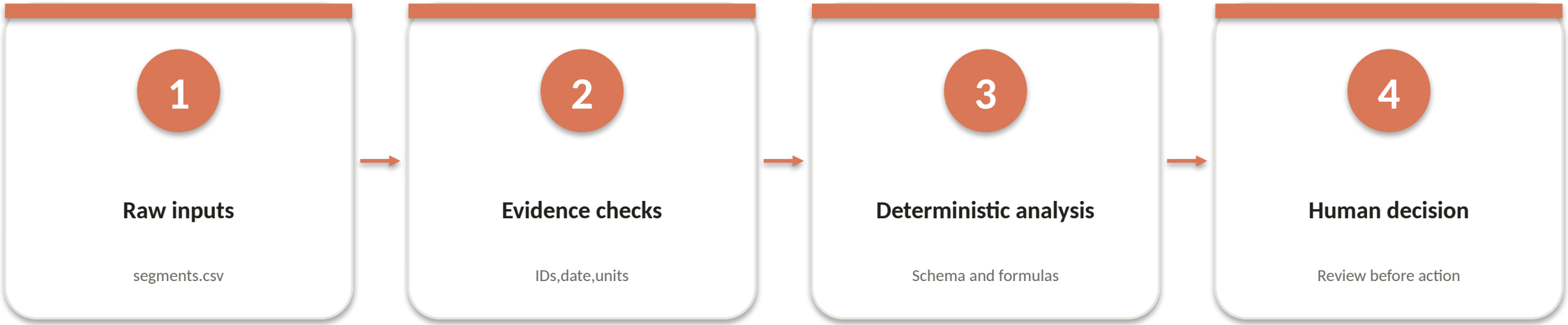
Claude Cowork · Claude Code · CSV / Markdown

DONE WHEN

Margins/customer 300,200,500; total customers 80; no invented demographic traits; retained-margin assumptions explicit.

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Segments and Customer Value · Data and Trust Boundaries



CONTRACT

Choose observable non-sensitive variables, compute group sizes and values, validate distinct needs with evidence, avoid assigning personas beyond available data.

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Segments and Customer Value · Input Schema

Field	Contract	Quality control
segment	str; example: Working professionals	Preserve segment;check units/identity
customers	int; example: 40	Preserve customers;check units/identity
revenue_sgd	int; example: 24000	Preserve revenue_sgd;check units/identity
cost_sgd	int; example: 12000	Preserve cost_sgd;check units/identity
retention	float; example: 0.7	Preserve retention;check units/identity

WHEN IT MATTERS Source rows remain immutable. New computed fields belong in outputs; ambiguous data is qualified rather than invented.

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Segments and Customer Value · Evidence Exhibit

segment	customers	revenue_sgd	cost_sgd
Working professionals	40	24000	12000
Career switchers	30	15000	9000
Team leads	10	8000	3000

WHEN IT MATTERS This exhibit is an authored synthetic dataset. Read exact units and identifiers before interpreting values.

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Segments and Customer Value · Worked Calculation

Interpret the denominator,units and decision boundary.

```
Working margin/customer = (24000-12000)/40 = $300.
Simple
one-period retained margin estimate = $300 × 0.7 =
$210;
not lifetime value.
```

Inputs

segments.csv

Calculation

Recompute exact values independently.

Interpretation

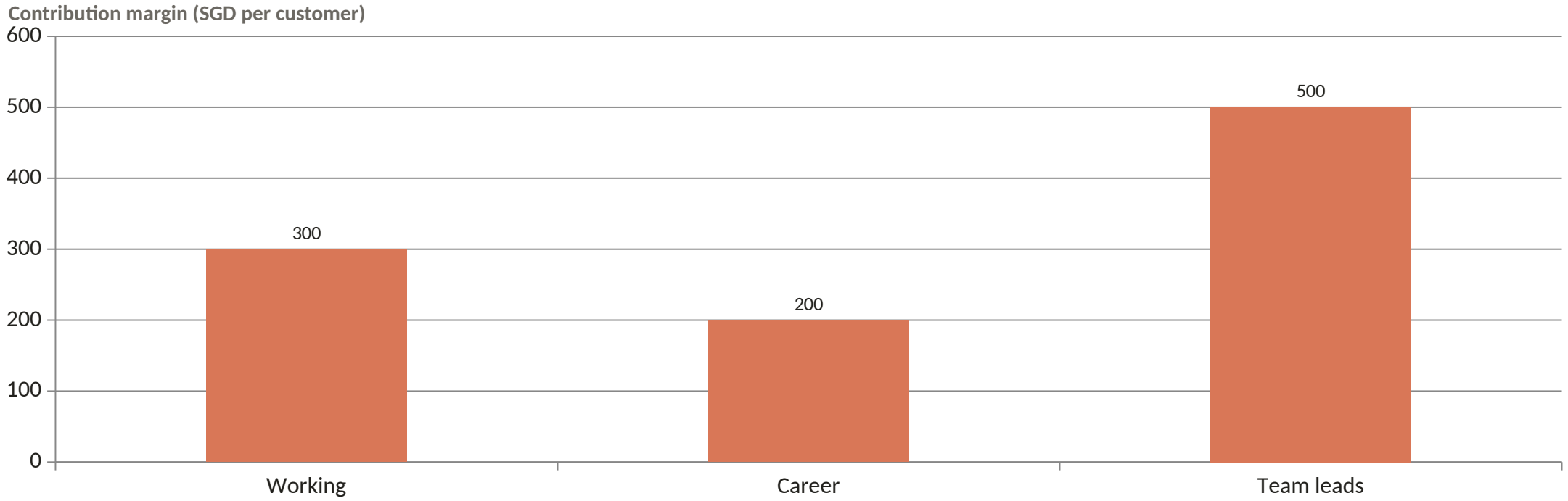
Synthetic contribution margin per customer. Team leads have higher unit margin but…

Boundary

Synthetic descriptive evidence;live/causal result not established.

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Segments and Customer Value · Editable Chart



WHAT THE DATA SHOWS

Synthetic contribution margin per customer. Team leads have higher unit margin but only ten customers; validate segment stability.

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Segments and Customer Value · Reproducible Contract

Readable excerpt; complete procedure and runnable practice remain in the guide.

```
import csv
with open("mock-data/segments.csv") as f:
    rows = list(csv.DictReader(f))
for r in rows:
    margin = float(r["revenue_sgd"])-
float(r["cost_sgd"])
    r["margin_per_customer"] =
margin/int(r["customers"])
```

Data contract

segments.csv; exact field names matter.

Identity

Preserve row IDs or segment labels.

Calculation

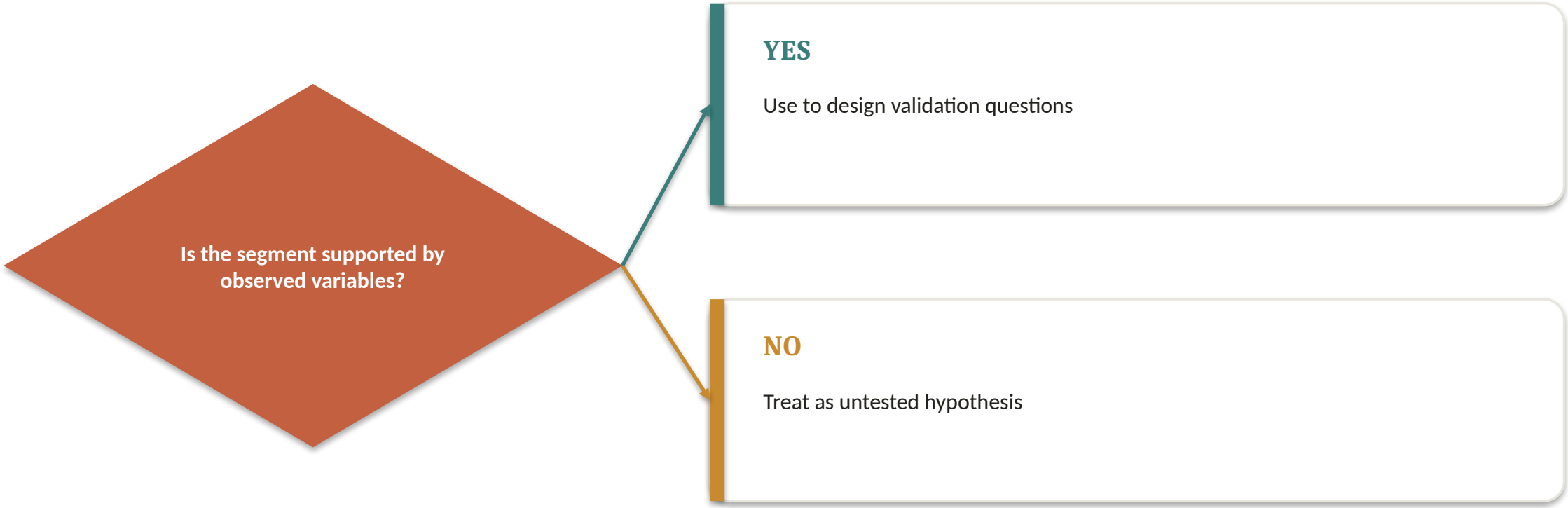
Deterministic code checks the agent narrative.

Verification

Margins/customer 300,200,500; total customers 80; no invented demographic traits…

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Segments and Customer Value · Decision Rule



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Segments and Customer Value · Claim-Evidence Ledger

Claim layer	Evidence required	Permitted wording
Observation	segment: Working professionals	Working margin/customer = (24000-12000)/40 = $300. Simple one-period retained margin estimate = $300 × 0.7 =…
Inference	Synthetic contribution margin per customer. Team leads have higher unit margin but only ten…	Is the segment supported by observed variables?
Recommendation	Use to design validation questions	persona-hypotheses.md
Unknown	The agent invents an age range and income for each segment. Neither field exists; the persona must remain…	Treat as untested hypothesis

WHEN IT MATTERS Preserve the distinction between observed data, model output and human decision throughout Cowork and Code handoffs.

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Segments and Customer Value · Failure Trace

A plausible report can be wrong even when its numbers look polished.

```
The agent invents an age range and income for each
segment. Neither field exists; the persona must remain
evidence-constrained.
```

Trigger

The agent promotes a claim beyond its evidence.

Diagnostic

Inspect the underlying rows,scope and denominator.

Recovery

Revise the claim and retain the rejected reasoning.

Recheck

Margins/customer 300,200,500; total customers 80; no invented demographic traits…

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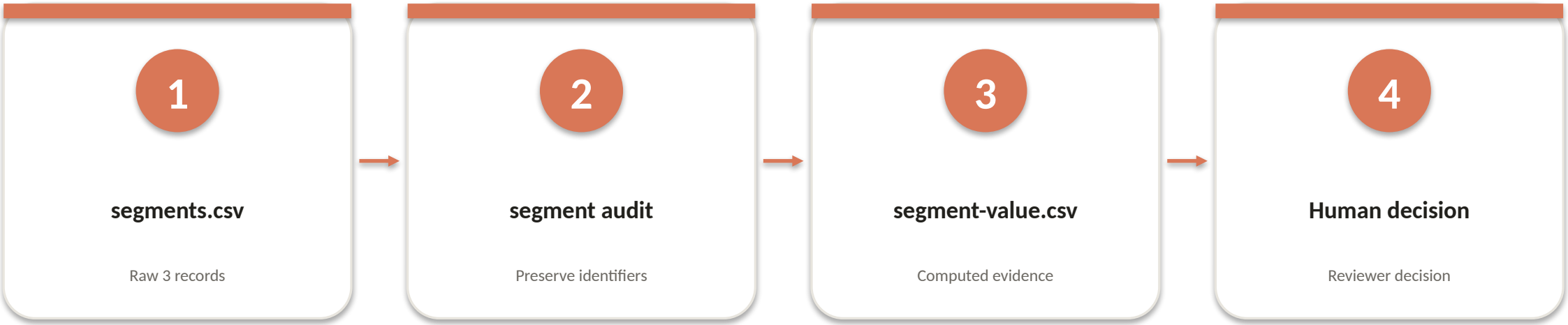
Segments and Customer Value · Control and Recovery

Control point	Risk	Required evidence
segment identity	Working professionals omitted or changed	Reconcile raw identifiers in segments.csv
Metric contract	Working margin/customer = (24000-12000)/40 = $300. Simple one-period retained margin estimate = $300 × 0.7 =…	Independent calculation and named denominator
Inference check	The agent invents an age range and income for each segment. Neither field exists; the persona must remain…	Treat as untested hypothesis
Decision output	segment-value.csv and persona-hypotheses.md	Use behavioural attributes and minimum group size for reporting; label retention as an assumption; suppress…

WHEN IT MATTERS Use behavioural attributes and minimum group size for reporting; label retention as an assumption; suppress identifying small-group combinations.

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Segments and Customer Value · Output Handoff



OUTPUT CONTRACT

Working margin/customer = (24000-12000)/40 = $300. Simple one-period retained margin estimate = $300 × 0.7 = $210; not lifetime value. Output: segment-value.csv and persona-hypotheses.md

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Segments and Customer Value · Acceptance Evidence

✅ Expected result

Margins/customer 300,200,500; total customers 80; no invented demographic traits; retained-margin assumptions explicit.

IF IT DOESN'T WORK

Numbers disagree

Check header parsing,units,denominator and duplicate records.

Claim lacks a source

Add row IDs and scope; downgrade or remove unsupported inference.

Tool unavailable

Use the documented CSV fallback and state the unexecuted product boundary.

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Journey and Funnel Diagnostics

ACTIVITY 06

Non-WSQ commercial short course · LO3

A synthetic monthly journey moves from landing sessions to enquiry, qualified lead and enrolment. The decision is where to investigate friction, using stage counts with consistent cohort and window definitions.

YOU'LL BUILD

funnel-analysis.csv and friction-hypotheses.md

TOOLCHAIN

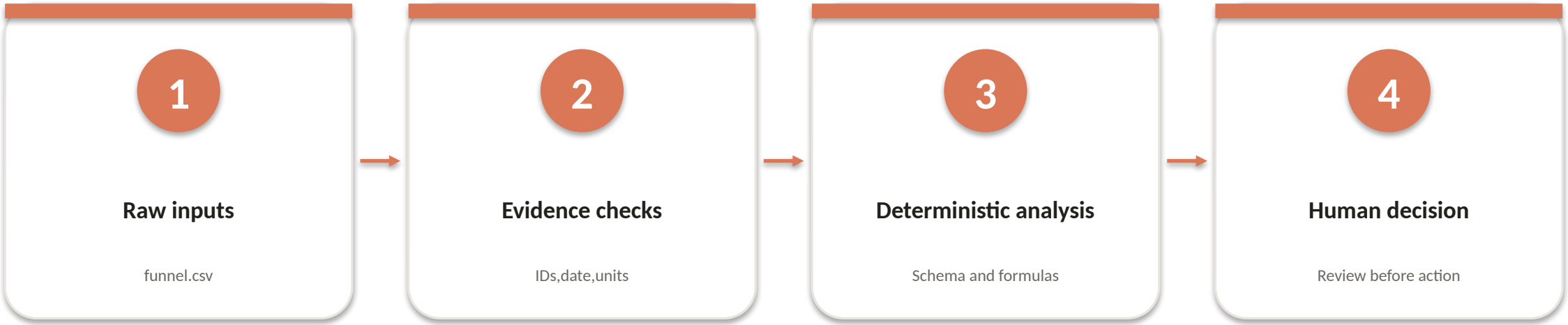
Claude Cowork · Claude Code · CSV / Markdown

DONE WHEN

Row-level journey-events.csv confirms2400 unique sessions,96 enquiries,48 qualified and12 enrolments;ordered stages share the same session IDs;rates 4%,50%,25%,0.5%;two alternate friction explanations.

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Journey and Funnel Diagnostics · Data and Trust Boundaries



CONTRACT

Use journey-events.csv to validate session-level ordered events in one synthetic cohort and30-day window;derive the aggregate stage counts,then compute conditional and overall rates. Raw aggregate counts alone do not establish a funnel.

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Journey and Funnel Diagnostics · Input Schema

Field	Contract	Quality control
stage	str; example: landing_session	Preserve stage;check units/identity
count	int; example: 2400	Preserve count;check units/identity
cohort	str; example: August visitors	Preserve cohort;check units/identity
window	str; example: 30 days	Preserve window;check units/identity

WHEN IT MATTERS Source rows remain immutable. New computed fields belong in outputs; ambiguous data is qualified rather than invented.

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Journey and Funnel Diagnostics · Evidence Exhibit

stage	count	cohort	window
landing_session	2400	August visitors	30 days
enquiry	96	August visitors	30 days
qualified_lead	48	August visitors	30 days
enrolment	12	August visitors	30 days

WHEN IT MATTERS This exhibit is an authored synthetic dataset. Read exact units and identifiers before interpreting values.

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Journey and Funnel Diagnostics · Worked Calculation

Interpret the denominator,units and decision boundary.

```
Session->enquiry = 96/2400 = 4%; enquiry->qualified =
50%; qualified->enrolment = 25%; full funnel = 0.5%.
```

Inputs

funnel.csv

Calculation

Recompute exact values independently.

Interpretation

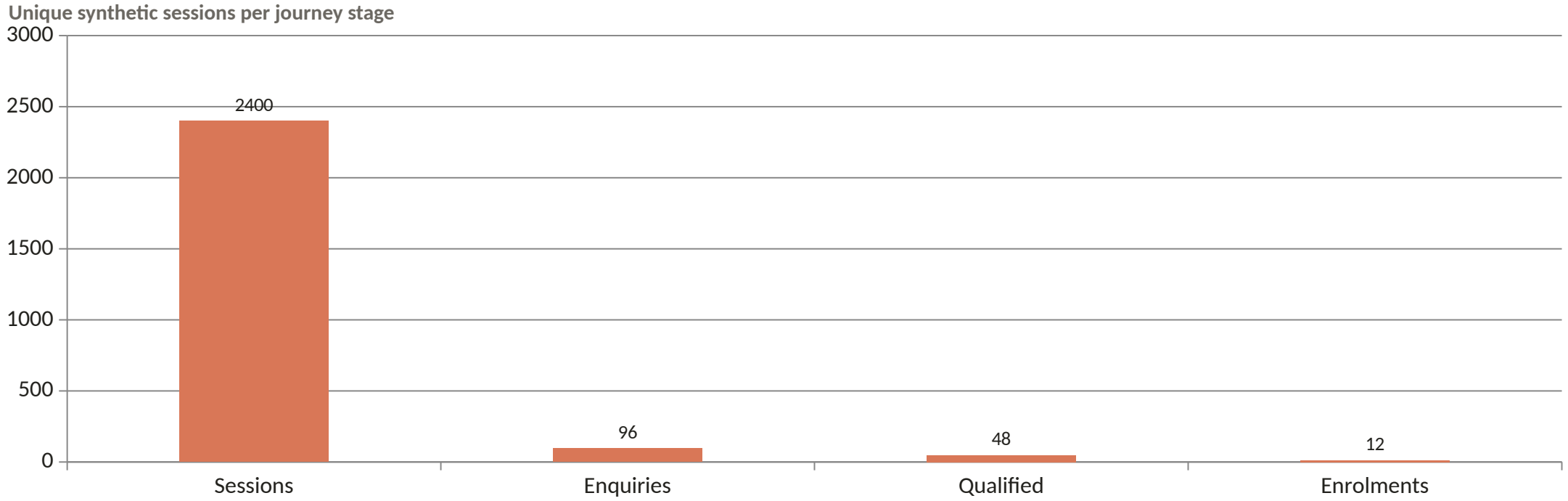
Synthetic counts show a large session-to-enquiry drop. It could reflect intent or…

Boundary

Synthetic descriptive evidence;live/causal result not established.

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Journey and Funnel Diagnostics · Editable Chart



WHAT THE DATA SHOWS

Synthetic counts show a large session-to-enquiry drop. It could reflect intent or tracking; the chart alone cannot identify the cause.

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Journey and Funnel Diagnostics · Reproducible Contract

Readable excerpt; complete procedure and runnable practice remain in the guide.

```
counts = [2400, 96, 48, 12]
rates = [b/a for a,b in zip(counts, counts[1:])]
assert rates == [0.04, 0.5, 0.25]
overall = counts[-1]/counts[0]
assert overall == 0.005
```

Data contract

funnel.csv; exact field names matter.

Identity

Preserve row IDs or segment labels.

Calculation

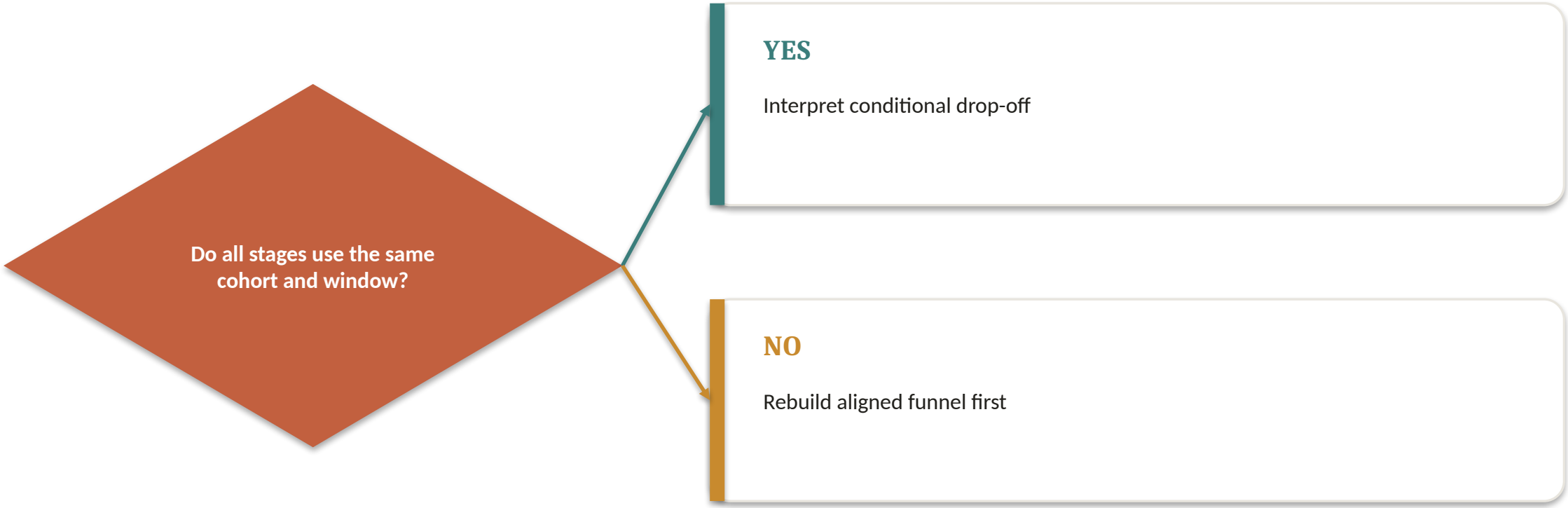
Deterministic code checks the agent narrative.

Verification

Row-level journey-events.csv confirms2400 unique sessions,96 enquiries,48 qualified…

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Journey and Funnel Diagnostics · Decision Rule



This is an authored research decision rule. It does not grant product permissions or replace human review.

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Journey and Funnel Diagnostics · Claim-Evidence Ledger

Claim layer	Evidence required	Permitted wording
Observation	stage: landing_session	Session->enquiry = 96/2400 = 4%; enquiry->qualified = 50%; qualified->enrolment = 25%; full funnel = 0.5%.
Inference	Synthetic counts show a large session-to-enquiry drop. It could reflect intent or tracking; the…	Do all stages use the same cohort and window?
Recommendation	Interpret conditional drop-off	friction-hypotheses.md
Unknown	A month-end enrolment count is divided by a different month's enquiries. Time-window mismatch invalidates the…	Rebuild aligned funnel first

WHEN IT MATTERS Preserve the distinction between observed data, model output and human decision throughout Cowork and Code handoffs.

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Journey and Funnel Diagnostics · Failure Trace

A plausible report can be wrong even when its numbers look polished.

```
A month-end enrolment count is divided by a different
month’s enquiries. Time-window mismatch invalidates
the
conversion rate.
```

Trigger

The agent promotes a claim beyond its evidence.

Diagnostic

Inspect the underlying rows,scope and denominator.

Recovery

Revise the claim and retain the rejected reasoning.

Recheck

Row-level journey-events.csv confirms2400 unique sessions,96 enquiries,48 qualified…

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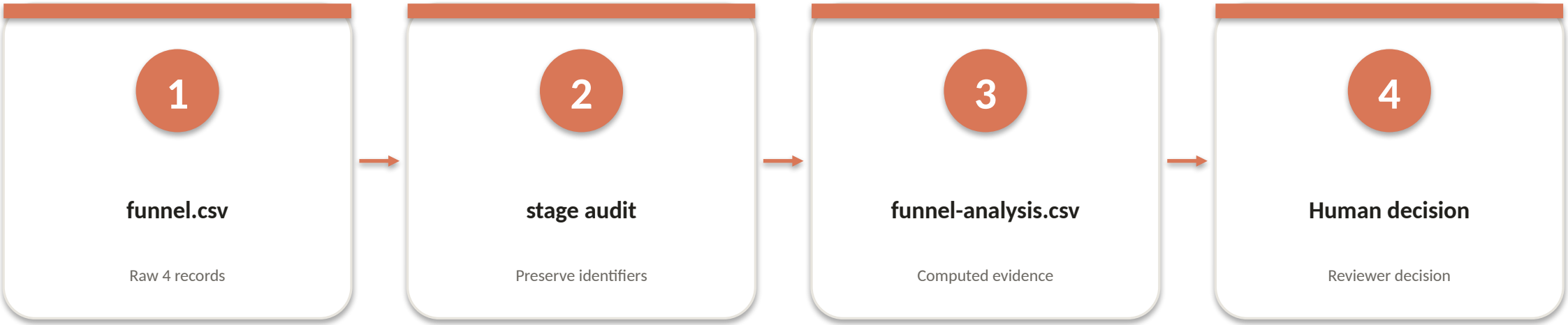
Journey and Funnel Diagnostics · Control and Recovery

Control point	Risk	Required evidence
stage identity	landing_session omitted or changed	Reconcile raw identifiers in funnel.csv
Metric contract	Session->enquiry = 96/2400 = 4%; enquiry->qualified = 50%; qualified->enrolment = 25%; full funnel = 0.5%.	Independent calculation and named denominator
Inference check	A month-end enrolment count is divided by a different month's enquiries. Time-window mismatch invalidates the…	Rebuild aligned funnel first
Decision output	funnel-analysis.csv and friction-hypotheses.md	State entity, cohort, order and window. Investigate consent and event completeness before changing creative…

WHEN IT MATTERS State entity, cohort, order and window. Investigate consent and event completeness before changing creative. Distinguish correlation from causal diagnosis.

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Journey and Funnel Diagnostics · Output Handoff



OUTPUT CONTRACT

Session->enquiry = 96/2400 = 4%; enquiry->qualified = 50%; qualified->enrolment = 25%; full funnel = 0.5%. Output: funnel-analysis.csv and friction-hypotheses.md

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Journey and Funnel Diagnostics · Acceptance Evidence

✅ Expected result

Row-level journey-events.csv confirms2400 unique sessions,96 enquiries,48 qualified and12 enrolments;ordered stages share the same session IDs;rates 4%,50%,25%,0.5%;two alternate friction explanations.

IF IT DOESN'T WORK

Numbers disagree

Check header parsing,units,denominator and duplicate records.

Claim lacks a source

Add row IDs and scope; downgrade or remove unsupported inference.

Tool unavailable

Use the documented CSV fallback and state the unexecuted product boundary.

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4

LEARNING OUTCOME 4

Marketing Effectiveness

Cowork evidence synthesis · Claude Code reproducible analysis

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GA4 Measurement Contract

ACTIVITY 07

Non-WSQ commercial short course · LO4

A marketing team needs GA4-compatible event definitions for course discovery, lead submission and purchase. Cowork can draft the measurement contract, but event collection and consent remain implementation responsibilities.

YOU'LL BUILD

measurement-contract.csv and consent-notes.md

TOOLCHAIN

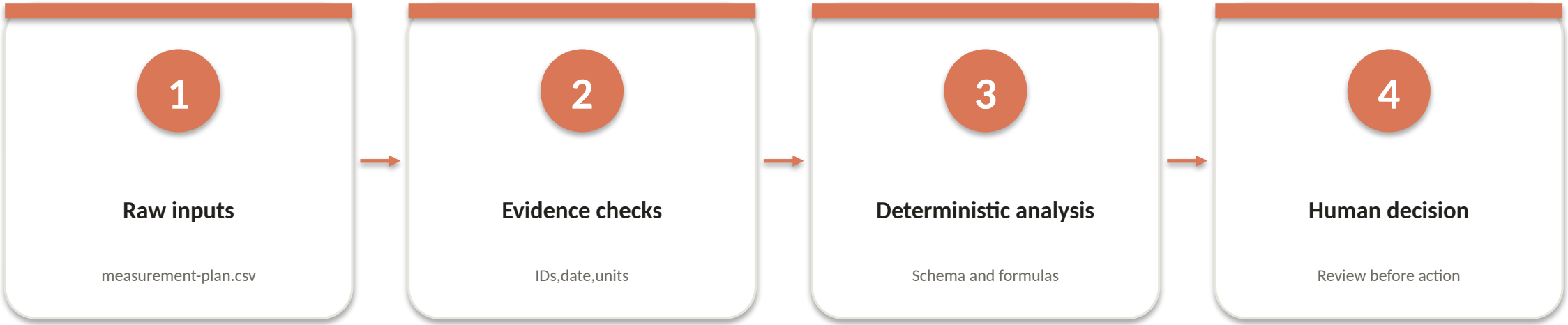
Claude Cowork · Claude Code · CSV / Markdown

DONE WHEN

Four events with unambiguous success triggers; purchase transaction_id,currency,value,items; no personal-data parameters.

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GA4 Measurement Contract · Data and Trust Boundaries



CONTRACT

Business question -> event trigger -> recommended event -> parameter schema -> consent -> validation evidence -> key-event designation. Avoid treating button clicks as successful submissions.

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GA4 Measurement Contract · Input Schema

Field	Contract	Quality control
event_name	str; example: page_view	Preserve event_name;check units/identity
trigger	str; example: page loads	Preserve trigger;check units/identity
required_parameter	str; example: page_location	Preserve required_parameter;check units/identity
purpose	str; example: discovery	Preserve purpose;check units/identity
key_event	str; example: no	Preserve key_event;check units/identity

WHEN IT MATTERS Source rows remain immutable. New computed fields belong in outputs; ambiguous data is qualified rather than invented.

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GA4 Measurement Contract · Evidence Exhibit

event_name	trigger	required_parameter
page_view	page loads	page_location
generate_lead	successful enquiry	value
purchase	confirmed payment	transaction_id
view_item	course detail viewed	items

WHEN IT MATTERS This exhibit is an authored synthetic dataset. Read exact units and identifiers before interpreting values.

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GA4 Measurement Contract · Worked Calculation

Interpret the denominator,units and decision boundary.

```
Contract completeness = 4 defined triggers / 4 events
=
100%; it does not establish live tracking coverage.
```

Inputs

measurement-plan.csv

Calculation

Recompute exact values independently.

Interpretation

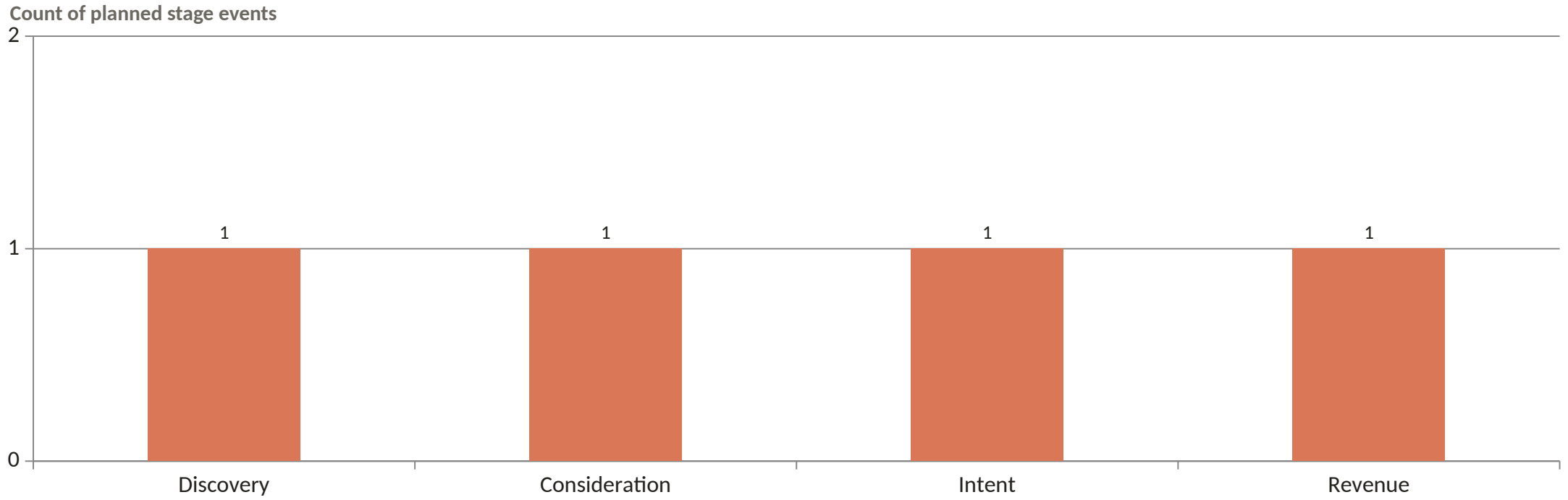
Synthetic plan covers four journey stages. These counts describe contract coverage…

Boundary

Synthetic descriptive evidence;live/causal result not established.

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GA4 Measurement Contract · Editable Chart



WHAT THE DATA SHOWS

Synthetic plan covers four journey stages. These counts describe contract coverage, not received GA4 traffic.

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GA4 Measurement Contract · Reproducible Contract

Readable excerpt; complete procedure and runnable practice remain in the guide.

```
# GA4 recommended purchase teaching payload
event = {"name": "purchase", "params": {
    "transaction_id": "SYN-T001",
    "currency": "SGD", "value": 600,
    "items": [{"item_id": "COURSE01",
               "item_name": "Synthetic programme"}]}}
```

Data contract

measurement-plan.csv; exact field names matter.

Identity

Preserve row IDs or segment labels.

Calculation

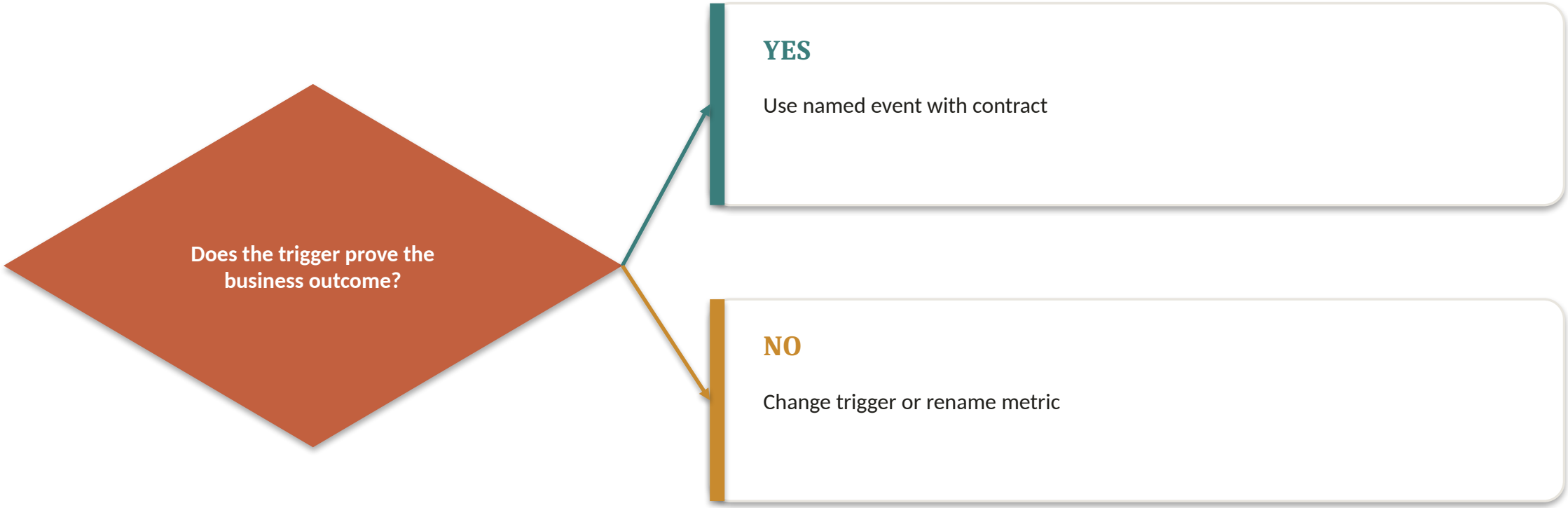
Deterministic code checks the agent narrative.

Verification

Four events with unambiguous success triggers; purchase…

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GA4 Measurement Contract · Decision Rule



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GA4 Measurement Contract · Claim-Evidence Ledger

Claim layer	Evidence required	Permitted wording
Observation	event_name: page_view	Contract completeness = 4 defined triggers / 4 events = 100%; it does not establish live tracking coverage.
Inference	Synthetic plan covers four journey stages. These counts describe contract coverage, not received…	Does the trigger prove the business outcome?
Recommendation	Use named event with contract	consent-notes.md
Unknown	generate_lead fires when a form button is pressed even if validation fails. The event trigger must be…	Change trigger or rename metric

WHEN IT MATTERS Preserve the distinction between observed data, model output and human decision throughout Cowork and Code handoffs.

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GA4 Measurement Contract · Failure Trace

A plausible report can be wrong even when its numbers look polished.

```
generate_lead fires when a form button is pressed even
if
validation fails. The event trigger must be confirmed
successful submission.
```

Trigger

The agent promotes a claim beyond its evidence.

Diagnostic

Inspect the underlying rows,scope and denominator.

Recovery

Revise the claim and retain the rejected reasoning.

Recheck

Four events with unambiguous success triggers; purchase…

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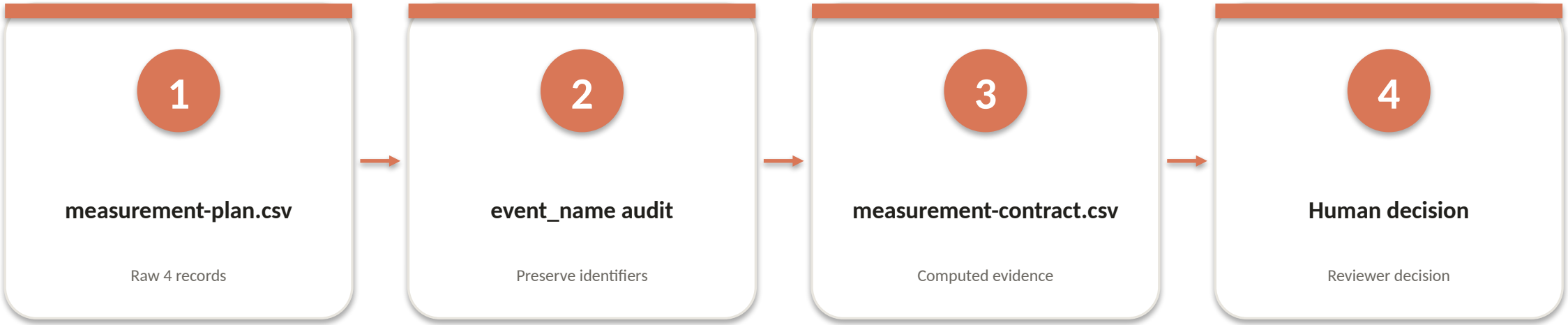
GA4 Measurement Contract · Control and Recovery

Control point	Risk	Required evidence
event_name identity	page_view omitted or changed	Reconcile raw identifiers in measurement-plan.csv
Metric contract	Contract completeness = 4 defined triggers / 4 events = 100%; it does not establish live tracking coverage.	Independent calculation and named denominator
Inference check	generate_lead fires when a form button is pressed even if validation fails. The event trigger must be…	Change trigger or rename metric
Decision output	measurement-contract.csv and consent-notes.md	Use recommended event semantics and current official documentation. Never send names, emails or phone numbers…

WHEN IT MATTERS Use recommended event semantics and current official documentation. Never send names, emails or phone numbers as analytics parameters. Collection is gated by the organisation’s consent policy.

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GA4 Measurement Contract · Output Handoff



OUTPUT CONTRACT

Contract completeness = 4 defined triggers / 4 events = 100%; it does not establish live tracking coverage. Output: measurement-contract.csv and consent-notes.md

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GA4 Measurement Contract · Acceptance Evidence

✅ Expected result

Four events with unambiguous success triggers; purchase transaction_id,currency,value,items; no personal-data parameters.

IF IT DOESN'T WORK

Numbers disagree

Check header parsing,units,denominator and duplicate records.

Claim lacks a source

Add row IDs and scope; downgrade or remove unsupported inference.

Tool unavailable

Use the documented CSV fallback and state the unexecuted product boundary.

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Event Quality and Deduplication

ACTIVITY 08

Non-WSQ commercial short course · LO4

A synthetic event export contains duplicate purchases and a missing transaction ID. Claude Code should create a reproducible audit without silently removing evidence or treating a CSV as a live GA4 test.

YOU'LL BUILD

event-audit.csv and reconciliation.md

TOOLCHAIN

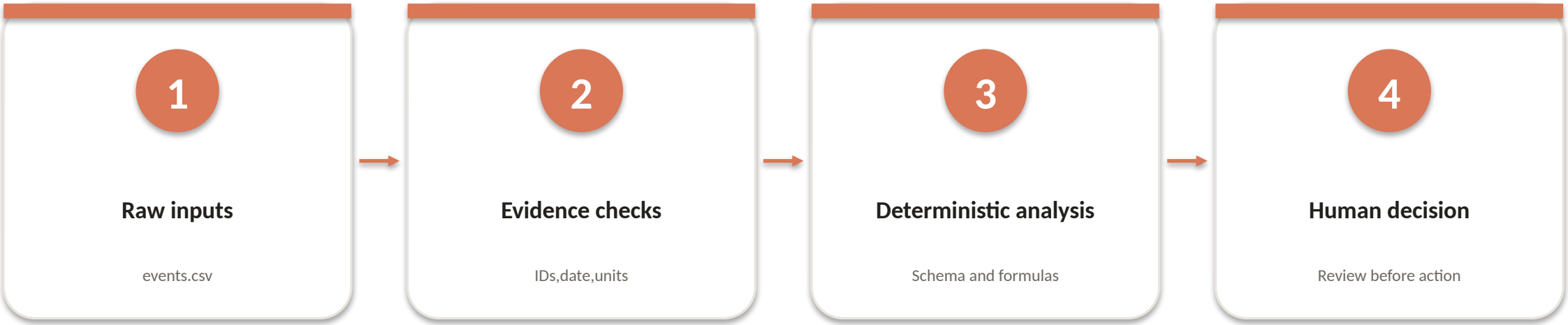
Claude Cowork · Claude Code · CSV / Markdown

DONE WHEN

Five raw rows retained in audit; T001 counted once; T002 once; E04 missing-ID; accepted value $1100.

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Event Quality and Deduplication · Data and Trust Boundaries



CONTRACT

Parse schema -> isolate purchases -> validate transaction IDs -> group duplicates -> reconcile value -> emit accepted and quarantined rows -> human review.

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Event Quality and Deduplication · Input Schema

Field	Contract	Quality control
event_id	str; example: E01	Preserve event_id;check units/identity
event_name	str; example: purchase	Preserve event_name;check units/identity
transaction_id	str; example: T001	Preserve transaction_id;check units/identity
value	int; example: 600	Preserve value;check units/identity
currency	str; example: SGD	Preserve currency;check units/identity

WHEN IT MATTERS Source rows remain immutable. New computed fields belong in outputs; ambiguous data is qualified rather than invented.

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Event Quality and Deduplication · Evidence Exhibit

event_id	transaction_id	value	currency
E01	T001	600	SGD
E02	T001	600	SGD
E03	T002	500	SGD
E04		400	SGD
E05		0	SGD

WHEN IT MATTERS This exhibit is an authored synthetic dataset. Read exact units and identifiers before interpreting values.

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Event Quality and Deduplication · Worked Calculation

Interpret the denominator,units and decision boundary.

```
Raw purchase value = 600+600+500+400 = $2100. Valid
distinct transaction value = $1100; duplicate $600 and
missing-ID $400 are quarantined.
```

Inputs

events.csv

Calculation

Recompute exact values independently.

Interpretation

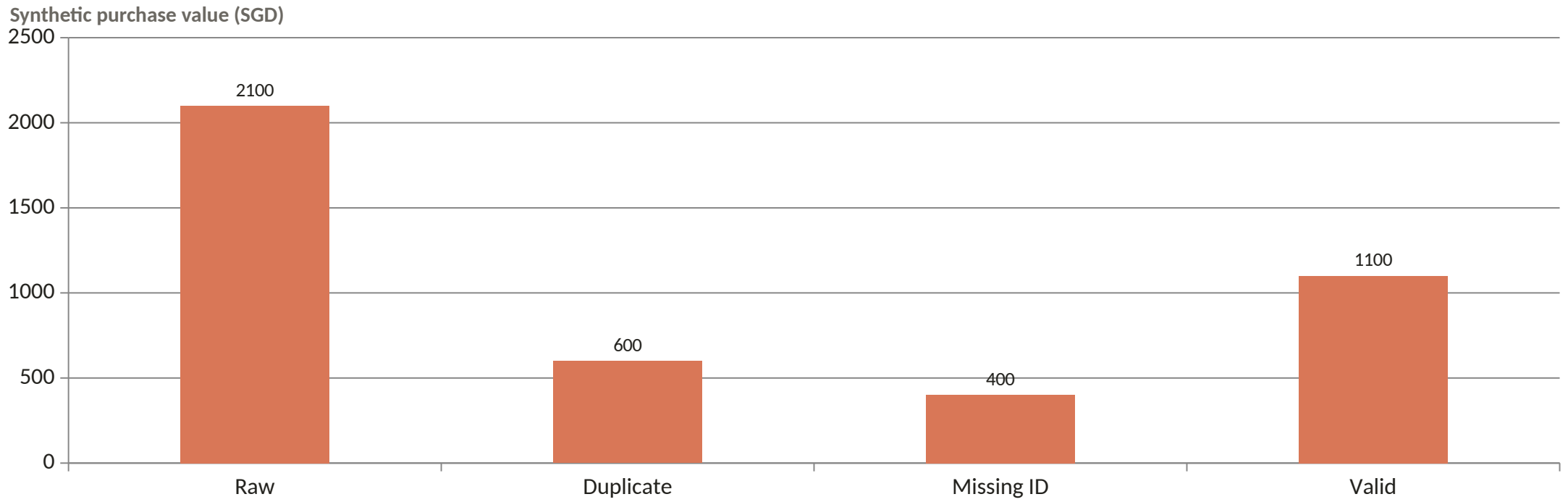
Synthetic reconciliation. Raw revenue is overstated by $1000; the audit must…

Boundary

Synthetic descriptive evidence;live/causal result not established.

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Event Quality and Deduplication · Editable Chart



WHAT THE DATA SHOWS

Synthetic reconciliation. Raw revenue is overstated by $1000; the audit must preserve excluded event IDs and reasons.

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Event Quality and Deduplication · Reproducible Contract

Readable excerpt; complete procedure and runnable practice remain in the guide.

```
purchases = [r for r in rows
             if r["event_name"] == "purchase"]
seen = set(); valid = []
for r in purchases:
    tid = r["transaction_id"]
    if tid and tid not in seen:
        valid.append(r); seen.add(tid)
```

Data contract

events.csv; exact field names matter.

Identity

Preserve row IDs or segment labels.

Calculation

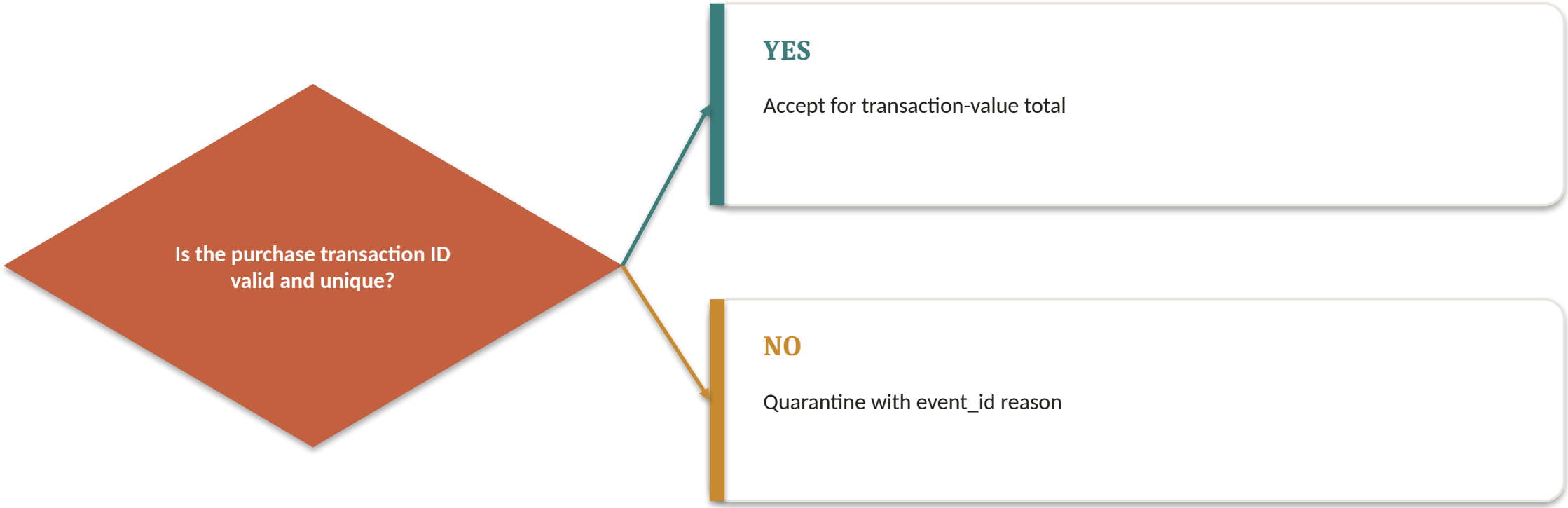
Deterministic code checks the agent narrative.

Verification

Five raw rows retained in audit; T001 counted once; T002 once; E04 missing-ID…

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Event Quality and Deduplication · Decision Rule



This is an authored research decision rule. It does not grant product permissions or replace human review.

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Event Quality and Deduplication · Claim-Evidence Ledger

Claim layer	Evidence required	Permitted wording
Observation	event_id: E01	Raw purchase value = 600+600+500+400 = $2100. Valid distinct transaction value = $1100; duplicate $600 and…
Inference	Synthetic reconciliation. Raw revenue is overstated by $1000; the audit must preserve excluded…	Is the purchase transaction ID valid and unique?
Recommendation	Accept for transaction-value total	reconciliation.md
Unknown	Deduplicating by value removes two different $600 orders. Transaction identity, not price equality…	Quarantine with event_id reason

WHEN IT MATTERS Preserve the distinction between observed data, model output and human decision throughout Cowork and Code handoffs.

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Event Quality and Deduplication · Failure Trace

A plausible report can be wrong even when its numbers look polished.

```
Deduplicating by value removes two different $600
orders.
Transaction identity, not price equality, determines
duplication.
```

Trigger

The agent promotes a claim beyond its evidence.

Diagnostic

Inspect the underlying rows,scope and denominator.

Recovery

Revise the claim and retain the rejected reasoning.

Recheck

Five raw rows retained in audit; T001 counted once; T002 once; E04 missing-ID…

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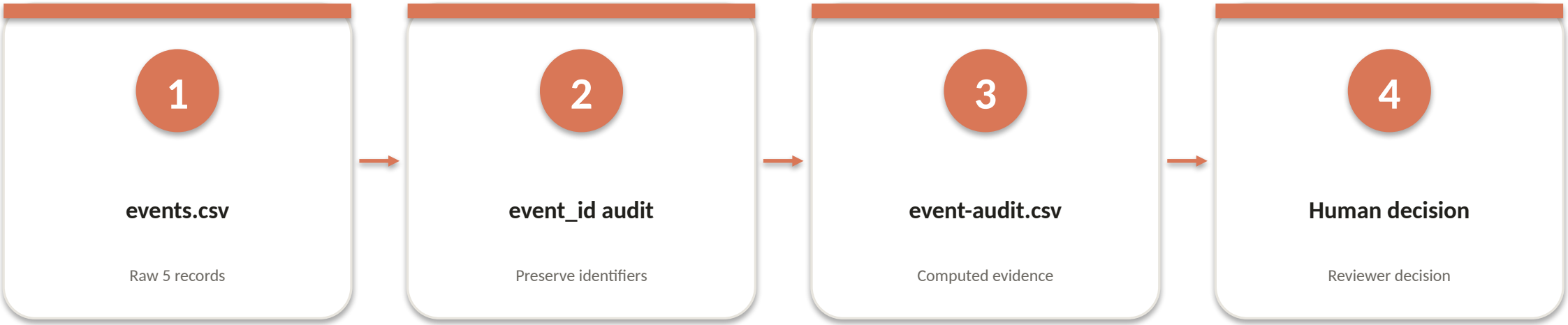
Event Quality and Deduplication · Control and Recovery

Control point	Risk	Required evidence
event_id identity	E01 omitted or changed	Reconcile raw identifiers in events.csv
Metric contract	Raw purchase value = 600+600+500+400 = $2100. Valid distinct transaction value = $1100; duplicate $600 and…	Independent calculation and named denominator
Inference check	Deduplicating by value removes two different $600 orders. Transaction identity, not price equality…	Quarantine with event_id reason
Decision output	event-audit.csv and reconciliation.md	Keep raw exports immutable, log rejection reasons, validate currency before summing, and make deduplication…

WHEN IT MATTERS Keep raw exports immutable, log rejection reasons, validate currency before summing, and make deduplication rules explicit. This dataset does not test actual GA4 deduplication.

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Event Quality and Deduplication · Output Handoff



OUTPUT CONTRACT

Raw purchase value = 600+600+500+400 = $2100. Valid distinct transaction value = $1100; duplicate $600 and missing-ID $400 are quarantined. Output: event-audit.csv and reconciliation.md

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Event Quality and Deduplication · Acceptance Evidence

✅ Expected result

Five raw rows retained in audit; T001 counted once; T002 once; E04 missing-ID; accepted value $1100.

IF IT DOESN'T WORK

Numbers disagree

Check header parsing,units,denominator and duplicate records.

Claim lacks a source

Add row IDs and scope; downgrade or remove unsupported inference.

Tool unavailable

Use the documented CSV fallback and state the unexecuted product boundary.

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Conversions and Attribution Limits

ACTIVITY 09

Non-WSQ commercial short course · LO4

Four synthetic conversion journeys include paid search, organic search, email and direct visits. Comparing first and last touch illustrates attribution sensitivity without pretending to reproduce GA4 data-driven attribution.

YOU'LL BUILD

attribution-comparison.csv and model-limitations.md

TOOLCHAIN

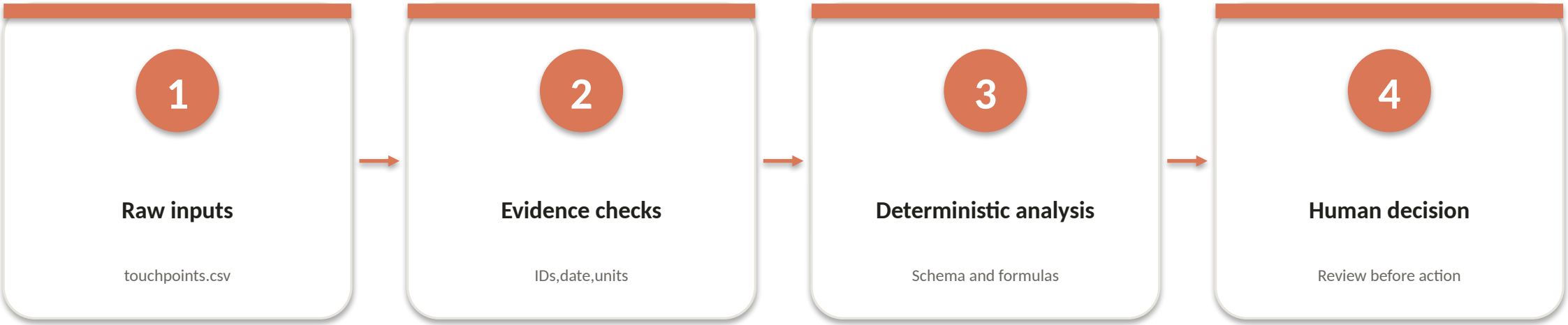
Claude Cowork · Claude Code · CSV / Markdown

DONE WHEN

Both models reconcile to $2400; paid first-touch $1200; email last-touch $1300; no causal ROI claim.

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Conversions and Attribution Limits · Data and Trust Boundaries



CONTRACT

Define eligible channels and lookback -> order touches -> choose allocation model -> reconcile totals -> compare sensitivity -> distinguish attribution from lift.

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Conversions and Attribution Limits · Input Schema

Field	Contract	Quality control
journey_id	str; example: J01	Preserve journey_id;check units/identity
first_channel	str; example: Paid search	Preserve first_channel;check units/identity
last_channel	str; example: Email	Preserve last_channel;check units/identity
revenue_sgd	int; example: 600	Preserve revenue_sgd;check units/identity

WHEN IT MATTERS Source rows remain immutable. New computed fields belong in outputs; ambiguous data is qualified rather than invented.

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Conversions and Attribution Limits · Evidence Exhibit

journey_id	first_channel	last_channel	revenue_sgd
J01	Paid search	Email	600
J02	Organic search	Direct	500
J03	Paid search	Direct	600
J04	Email	Email	700

WHEN IT MATTERS This exhibit is an authored synthetic dataset. Read exact units and identifiers before interpreting values.

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Conversions and Attribution Limits · Worked Calculation

Interpret the denominator,units and decision boundary.

```
First-touch paid search revenue = $1200; last-touch =
$0.
Email receives $700 first-touch and $1300 last-touch.
Model choice changes assigned credit.
```

Inputs

touchpoints.csv

Calculation

Recompute exact values independently.

Interpretation

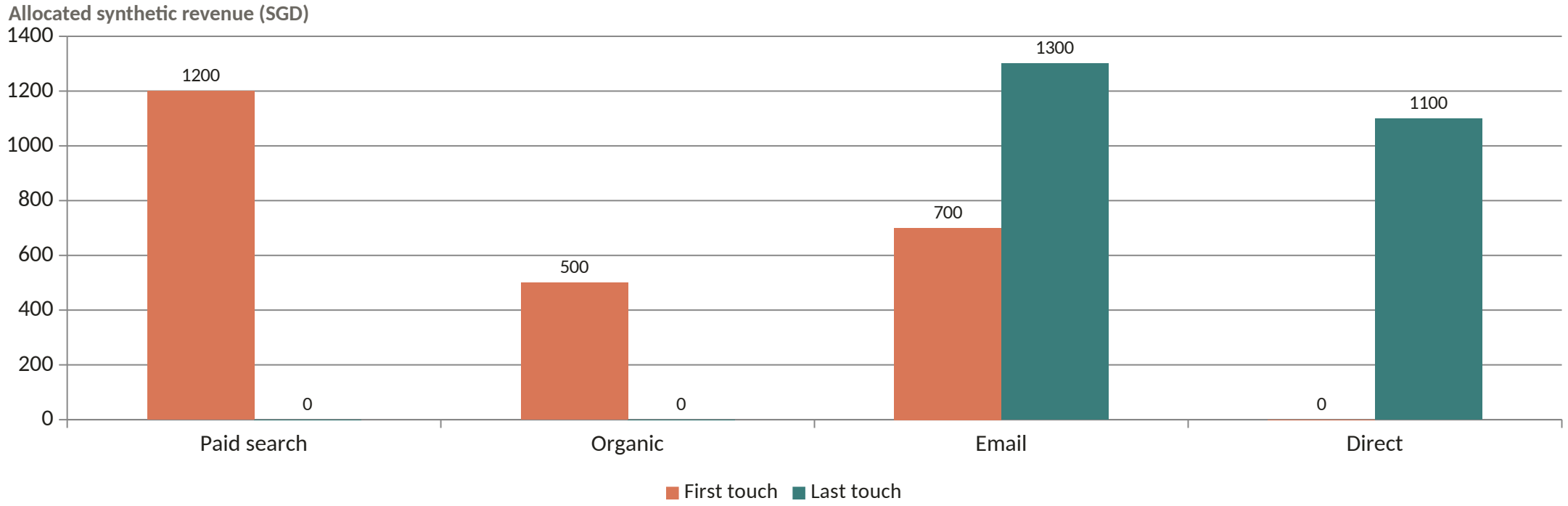
Synthetic first/last-touch allocation, both total $2400. Neither is an estimate of…

Boundary

Synthetic descriptive evidence;live/causal result not established.

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Conversions and Attribution Limits · Editable Chart



WHAT THE DATA SHOWS

Synthetic first/last-touch allocation, both total $2400. Neither is an estimate of incremental causal impact or GA4 data-driven credit.

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Conversions and Attribution Limits · Reproducible Contract

Readable excerpt; complete procedure and runnable practice remain in the guide.

```
from collections import defaultdict
first = defaultdict(float); last = defaultdict(float)
for r in rows:
    v = float(r["revenue_sgd"])
    first[r["first_channel"]] += v
    last[r["last_channel"]] += v
```

Data contract

touchpoints.csv; exact field names matter.

Identity

Preserve row IDs or segment labels.

Calculation

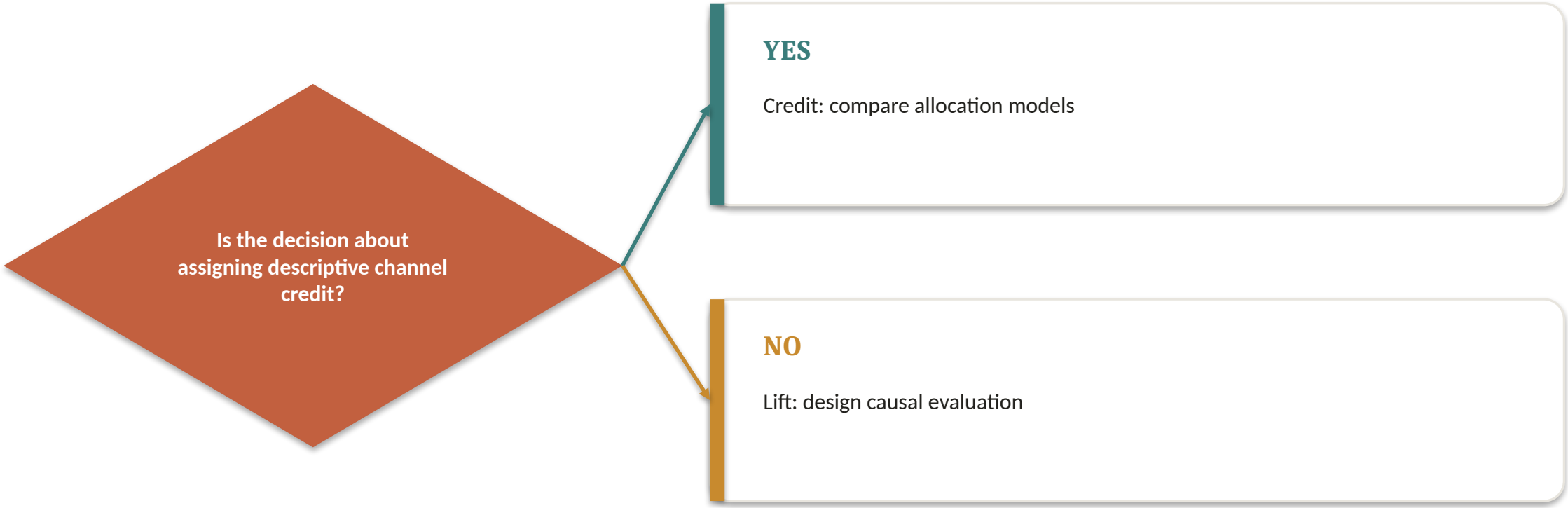
Deterministic code checks the agent narrative.

Verification

Both models reconcile to $2400; paid first-touch $1200; email last-touch $1300; no…

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Conversions and Attribution Limits · Decision Rule



This is an authored research decision rule. It does not grant product permissions or replace human review.

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Conversions and Attribution Limits · Claim-Evidence Ledger

Claim layer	Evidence required	Permitted wording
Observation	journey_id: J01	First-touch paid search revenue = $1200; last-touch = $0. Email receives $700 first-touch and $1300…
Inference	Synthetic first/last-touch allocation, both total $2400. Neither is an estimate of incremental…	Is the decision about assigning descriptive channel credit?
Recommendation	Credit: compare allocation models	model-limitations.md
Unknown	The agent cuts paid search because last-touch credit is zero. Assisted demand and incremental impact require…	Lift: design causal evaluation

WHEN IT MATTERS Preserve the distinction between observed data, model output and human decision throughout Cowork and Code handoffs.

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Conversions and Attribution Limits · Failure Trace

A plausible report can be wrong even when its numbers look polished.

```
The agent cuts paid search because last-touch credit
is
zero. Assisted demand and incremental impact require
separate investigation.
```

Trigger

The agent promotes a claim beyond its evidence.

Diagnostic

Inspect the underlying rows,scope and denominator.

Recovery

Revise the claim and retain the rejected reasoning.

Recheck

Both models reconcile to $2400; paid first-touch $1200; email last-touch $1300; no…

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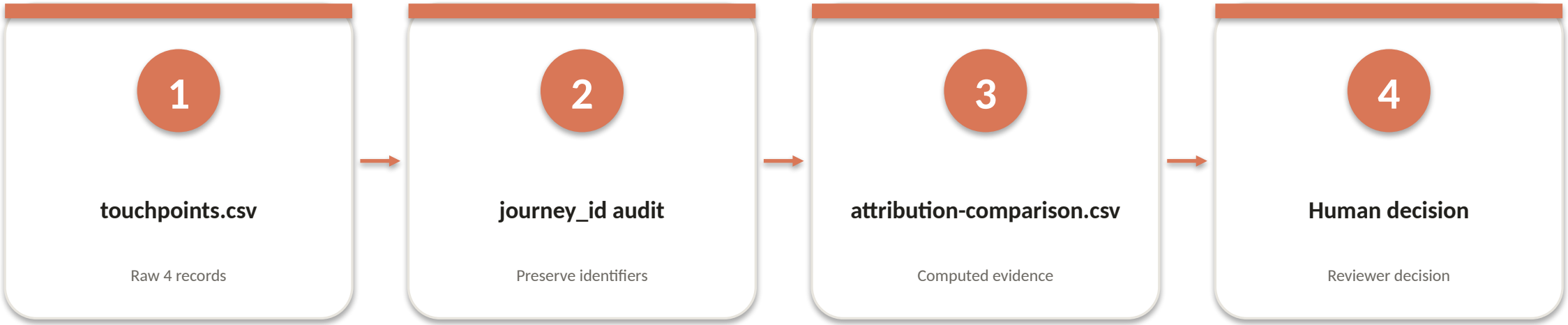
Conversions and Attribution Limits · Control and Recovery

Control point	Risk	Required evidence
journey_id identity	J01 omitted or changed	Reconcile raw identifiers in touchpoints.csv
Metric contract	First-touch paid search revenue = $1200; last-touch = $0. Email receives $700 first-touch and $1300…	Independent calculation and named denominator
Inference check	The agent cuts paid search because last-touch credit is zero. Assisted demand and incremental impact require…	Lift: design causal evaluation
Decision output	attribution-comparison.csv and model-limitations.md	Use attribution for descriptive allocation; use controlled experiments or credible causal designs for…

WHEN IT MATTERS Use attribution for descriptive allocation; use controlled experiments or credible causal designs for incremental effects. Explicitly state identity, consent and lookback limitations.

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Conversions and Attribution Limits · Output Handoff



OUTPUT CONTRACT

First-touch paid search revenue = $1200; last-touch = $0. Email receives $700 first-touch and $1300 last-touch. Model choice changes assigned credit. Output: attribution-comparison.csv and model-limitations.md

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Conversions and Attribution Limits · Acceptance Evidence

✅ Expected result

Both models reconcile to $2400; paid first-touch $1200; email last-touch $1300; no causal ROI claim.

IF IT DOESN'T WORK

Numbers disagree

Check header parsing,units,denominator and duplicate records.

Claim lacks a source

Add row IDs and scope; downgrade or remove unsupported inference.

Tool unavailable

Use the documented CSV fallback and state the unexecuted product boundary.

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3

LEARNING OUTCOME 3

Behaviour and Demand

Cowork evidence synthesis · Claude Code reproducible analysis

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Lifecycle Creative from Evidence

ACTIVITY 10

Non-WSQ commercial short course · LO3

A fictional email campaign targets enquiry and enrolled segments. Cowork must adapt evidence-backed messages to a brand brief while preserving consent rules, avoiding fabricated urgency and producing drafts for human review.

YOU'LL BUILD

message-variants.md and review-checklist.csv

TOOLCHAIN

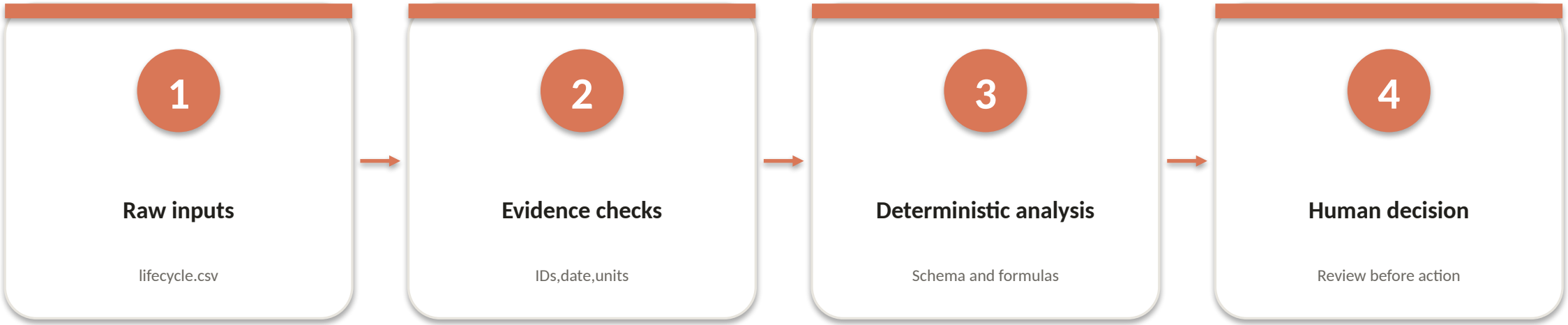
Claude Cowork · Claude Code · CSV / Markdown

DONE WHEN

Three segment drafts; claim IDs and CTA; no invented scarcity; send_authorised false; rates use sent.

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Lifecycle Creative from Evidence · Data and Trust Boundaries



CONTRACT

Evidence brief -> segment need -> approved claim -> channel/consent constraint -> draft variant -> human brand review -> measurement hypothesis.

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Lifecycle Creative from Evidence · Input Schema

Field	Contract	Quality control
segment	str; example: New enquiry	Preserve segment;check units/identity
eligible	int; example: 100	Preserve eligible;check units/identity
sent	int; example: 80	Preserve sent;check units/identity
clicked	int; example: 16	Preserve clicked;check units/identity
enrolled	int; example: 4	Preserve enrolled;check units/identity

WHEN IT MATTERS Source rows remain immutable. New computed fields belong in outputs; ambiguous data is qualified rather than invented.

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Lifecycle Creative from Evidence · Evidence Exhibit

segment	sent	clicked	enrolled
New enquiry	80	16	4
Qualified lead	40	12	6
Past learner	90	18	9

WHEN IT MATTERS This exhibit is an authored synthetic dataset. Read exact units and identifiers before interpreting values.

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Lifecycle Creative from Evidence · Worked Calculation

Interpret the denominator,units and decision boundary.

```
Click rate uses sent: 16/80=20%,12/40=30%,18/90=20%.
Enrolment per sent: 5%,15%,10%; no claim of
incremental
email impact.
```

Inputs

lifecycle.csv

Calculation

Recompute exact values independently.

Interpretation

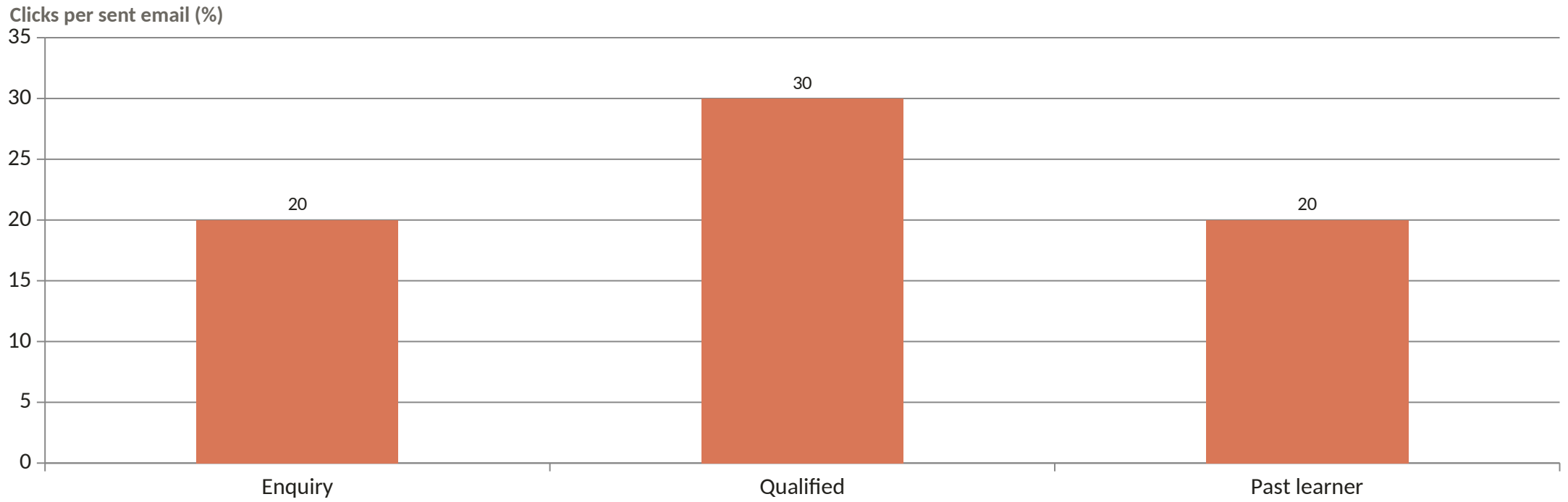
Synthetic click rates by sent denominator. Qualified leads respond more, but…

Boundary

Synthetic descriptive evidence;live/causal result not established.

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Lifecycle Creative from Evidence · Editable Chart



WHAT THE DATA SHOWS

Synthetic click rates by sent denominator. Qualified leads respond more, but eligibility and existing intent differ across segments.

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Lifecycle Creative from Evidence · Reproducible Contract

Readable excerpt; complete procedure and runnable practice remain in the guide.

```
# Authored lifecycle output contract
draft = {"segment": "Qualified lead",
 "claim_ids": ["CL01"], "status": "human_review",
 "subject": "Compare evening learning options",
 "cta": "View verified course details",
 "send_authorised": False}
```

Data contract

lifecycle.csv; exact field names matter.

Identity

Preserve row IDs or segment labels.

Calculation

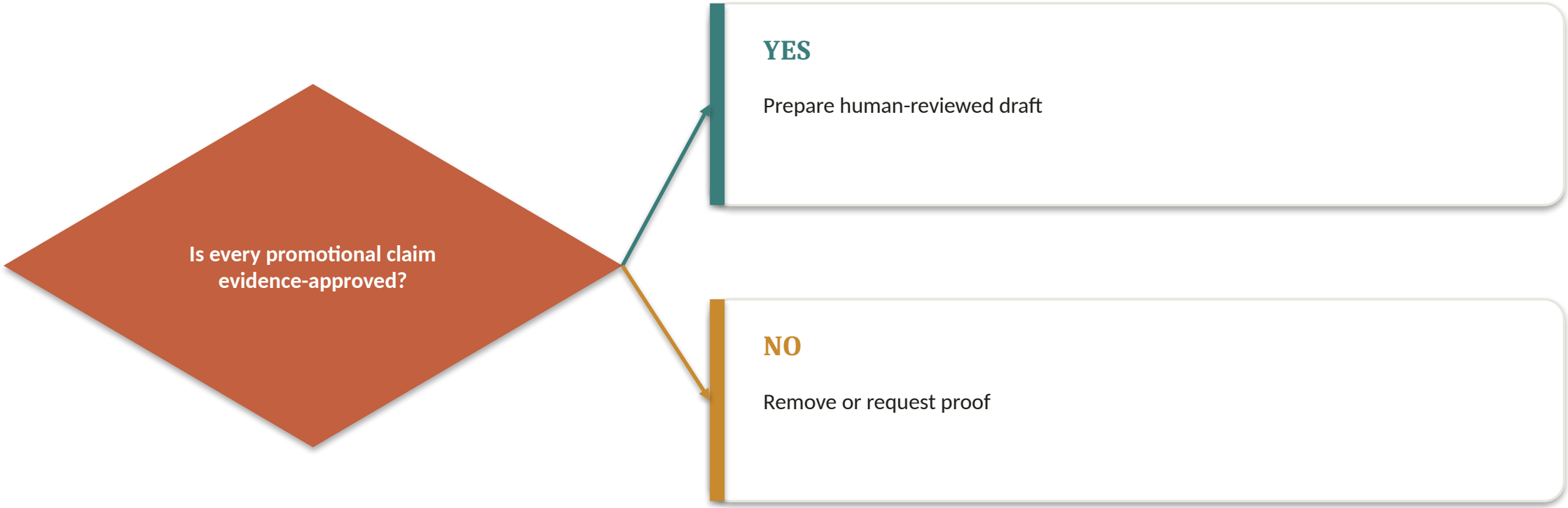
Deterministic code checks the agent narrative.

Verification

Three segment drafts; claim IDs and CTA; no invented scarcity; send_authorised…

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Lifecycle Creative from Evidence · Decision Rule



This is an authored research decision rule. It does not grant product permissions or replace human review.

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Lifecycle Creative from Evidence · Claim-Evidence Ledger

Claim layer	Evidence required	Permitted wording
Observation	segment: New enquiry	Click rate uses sent: 16/80=20%,12/40=30%,18/90=20%. Enrolment per sent: 5%,15%,10%; no claim of incremental…
Inference	Synthetic click rates by sent denominator. Qualified leads respond more, but eligibility and…	Is every promotional claim evidence-approved?
Recommendation	Prepare human-reviewed draft	review-checklist.csv
Unknown	A draft says only two seats remain without inventory evidence. The urgency claim must be removed or verified…	Remove or request proof

WHEN IT MATTERS Preserve the distinction between observed data, model output and human decision throughout Cowork and Code handoffs.

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Lifecycle Creative from Evidence · Failure Trace

A plausible report can be wrong even when its numbers look polished.

```
A draft says only two seats remain without inventory
evidence. The urgency claim must be removed or
verified
by an authorised source.
```

Trigger

The agent promotes a claim beyond its evidence.

Diagnostic

Inspect the underlying rows,scope and denominator.

Recovery

Revise the claim and retain the rejected reasoning.

Recheck

Three segment drafts; claim IDs and CTA; no invented scarcity; send_authorised…

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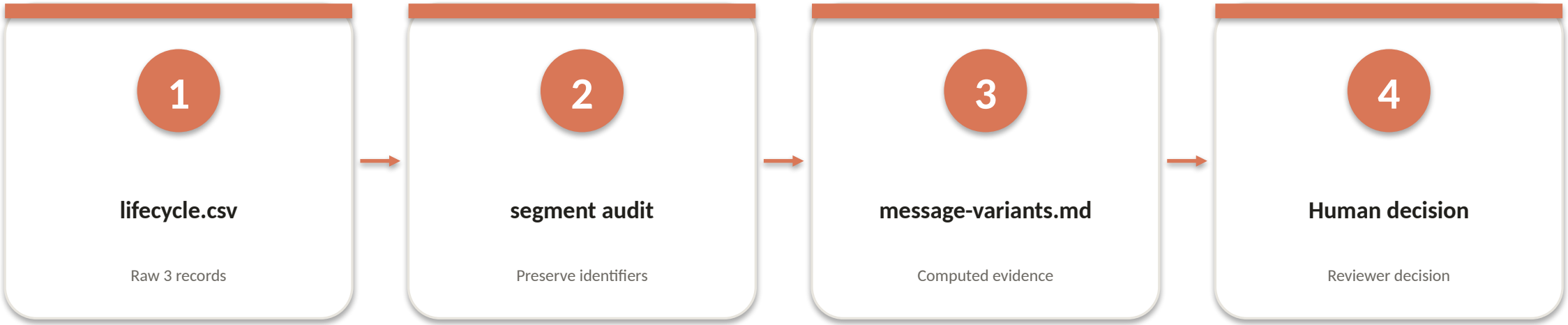
Lifecycle Creative from Evidence · Control and Recovery

Control point	Risk	Required evidence
segment identity	New enquiry omitted or changed	Reconcile raw identifiers in lifecycle.csv
Metric contract	Click rate uses sent: 16/80=20%,12/40=30%,18/90=20%. Enrolment per sent: 5%,15%,10%; no claim of incremental…	Independent calculation and named denominator
Inference check	A draft says only two seats remain without inventory evidence. The urgency claim must be removed or verified…	Remove or request proof
Decision output	message-variants.md and review-checklist.csv	Draft only; no send authorisation. Use approved factual claims, consented audiences, clear opt-out and…

WHEN IT MATTERS Draft only; no send authorisation. Use approved factual claims, consented audiences, clear opt-out and reviewer sign-off. Avoid interpreting synthetic clicks as causal lift.

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Lifecycle Creative from Evidence · Output Handoff



OUTPUT CONTRACT

Click rate uses sent: 16/80=20%,12/40=30%,18/90=20%. Enrolment per sent: 5%,15%,10%; no claim of incremental email impact. Output: message-variants.md and review-checklist.csv

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Lifecycle Creative from Evidence · Acceptance Evidence

✅ Expected result

Three segment drafts; claim IDs and CTA; no invented scarcity; send_authorised false; rates use sent.

IF IT DOESN'T WORK

Numbers disagree

Check header parsing,units,denominator and duplicate records.

Claim lacks a source

Add row IDs and scope; downgrade or remove unsupported inference.

Tool unavailable

Use the documented CSV fallback and state the unexecuted product boundary.

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2

LEARNING OUTCOME 2

Trends and Variables

Cowork evidence synthesis · Claude Code reproducible analysis

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SEO Content Experiment

ACTIVITY 11

Non-WSQ commercial short course · LO2

A synthetic search-content export tracks impressions, clicks and enquiries. Cowork should cluster intent and propose an experiment, using observed rate information without claiming an invented search ranking or guaranteed AI visibility.

YOU'LL BUILD

intent-map.csv and experiment-brief.md

TOOLCHAIN

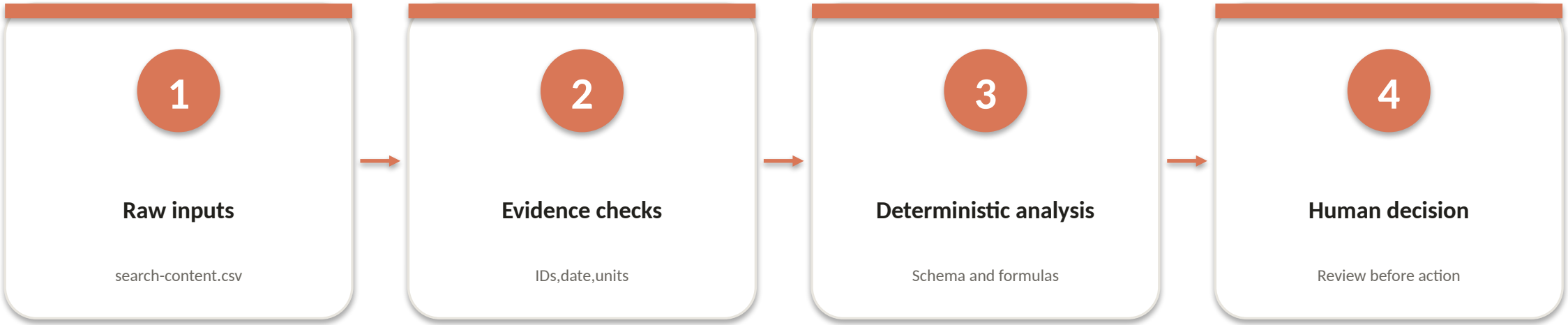
Claude Cowork · Claude Code · CSV / Markdown

DONE WHEN

CTRs 4%,3%,6%,3%; training enquiry/click 12.5%; one isolated change and explicit confounder.

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SEO Content Experiment · Data and Trust Boundaries



CONTRACT

Normalise query -> classify intent -> map page/claim -> calculate CTR and enquiry rates -> isolate one content change -> define control/window -> assess uncertainty.

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SEO Content Experiment · Input Schema

Field	Contract	Quality control
query	str; example: ai marketing course	Preserve query;check units/identity
impressions	int; example: 1000	Preserve impressions;check units/identity
clicks	int; example: 40	Preserve clicks;check units/identity
enquiries	int; example: 4	Preserve enquiries;check units/identity
intent	str; example: commercial	Preserve intent;check units/identity

WHEN IT MATTERS Source rows remain immutable. New computed fields belong in outputs; ambiguous data is qualified rather than invented.

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SEO Content Experiment · Evidence Exhibit

query	impressions	clicks	enquiries
ai marketing course	1000	40	4
claude cowork guide	2000	60	3
market research training	800	48	6
email segmentation examples	1200	36	2

WHEN IT MATTERS This exhibit is an authored synthetic dataset. Read exact units and identifiers before interpreting values.

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SEO Content Experiment · Worked Calculation

Interpret the denominator,units and decision boundary.

```
CTR = clicks/impressions. Training query: 48/800=6%;
enquiry/click = 6/48=12.5%. High impression volume
alone
does not establish useful demand.
```

Inputs

search-content.csv

Calculation

Recompute exact values independently.

Interpretation

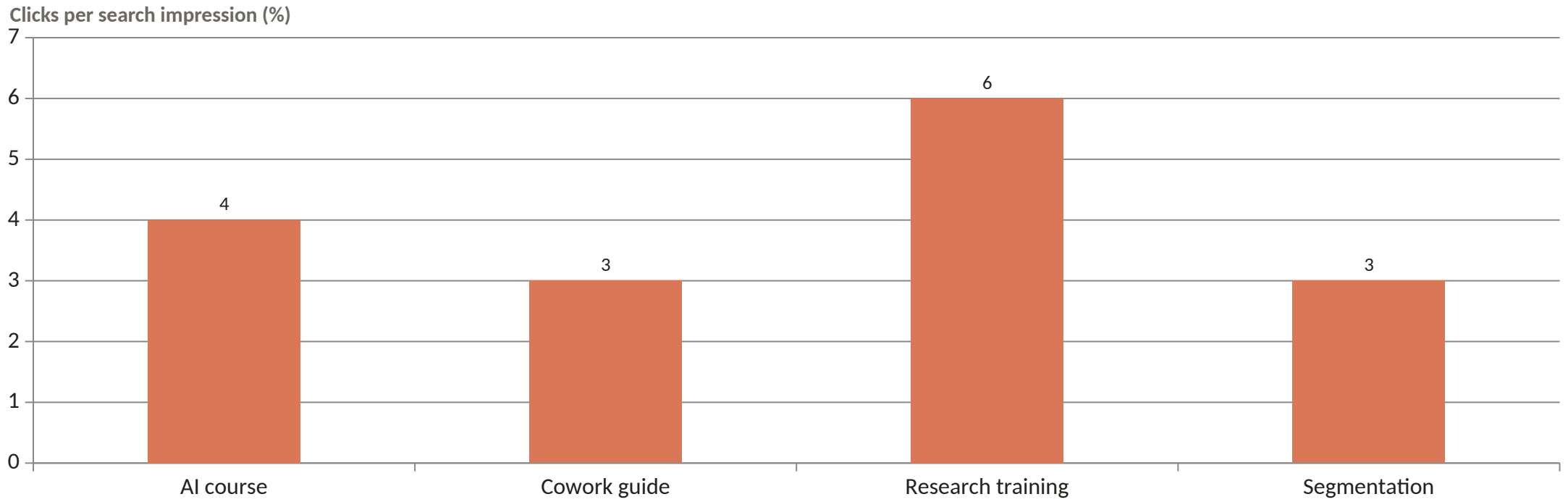
Synthetic CTR percentages. Research training has higher CTR and enquiry rate; intent...

Boundary

Synthetic descriptive evidence;live/causal result not established.

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SEO Content Experiment · Editable Chart



WHAT THE DATA SHOWS

Synthetic CTR percentages. Research training has higher CTR and enquiry rate; intent and page relevance are hypotheses to validate.

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SEO Content Experiment · Reproducible Contract

Readable excerpt; complete procedure and runnable practice remain in the guide.

```
for r in rows:
    impressions = int(r["impressions"])
    clicks = int(r["clicks"])
    r["ctr"] = clicks/impressions
    r["enquiry_per_click"] = (
        int(r["enquiries"])/clicks if clicks else 0)
```

Data contract

search-content.csv; exact field names matter.

Identity

Preserve row IDs or segment labels.

Calculation

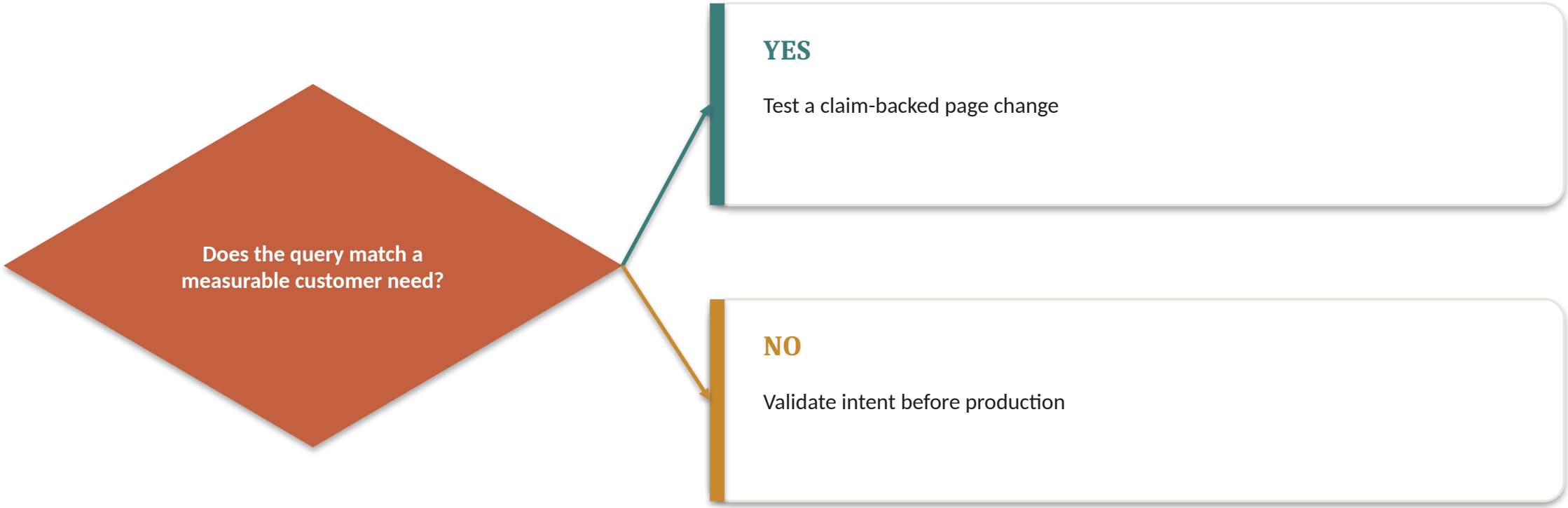
Deterministic code checks the agent narrative.

Verification

CTRs 4%,3%,6%,3%; training enquiry/click 12.5%; one isolated change and explicit…

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SEO Content Experiment · Decision Rule



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SEO Content Experiment · Claim-Evidence Ledger

Claim layer	Evidence required	Permitted wording
Observation	query: ai marketing course	CTR = clicks/impressions. Training query: 48/800=6%; enquiry/click = 6/48=12.5%. High impression volume alone…
Inference	Synthetic CTR percentages. Research training has higher CTR and enquiry rate; intent and page…	Does the query match a measurable customer need?
Recommendation	Test a claim-backed page change	experiment-brief.md
Unknown	A content plan targets the highest-impression query and ignores enquiry quality. Optimise against the…	Validate intent before production

WHEN IT MATTERS Preserve the distinction between observed data, model output and human decision throughout Cowork and Code handoffs.

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LO2 · ACTIVITY 11 · SYNTHETIC TEACHING DATA

SEO Content Experiment · Failure Trace

A plausible report can be wrong even when its numbers look polished.

```
A content plan targets the highest-impression query
and
ignores enquiry quality. Optimise against the decision
metric rather than raw reach.
```

Trigger

The agent promotes a claim beyond its evidence.

Diagnostic

Inspect the underlying rows,scope and denominator.

Recovery

Revise the claim and retain the rejected reasoning.

Recheck

CTRs 4%,3%,6%,3%; training enquiry/click 12.5%; one isolated change and explicit…

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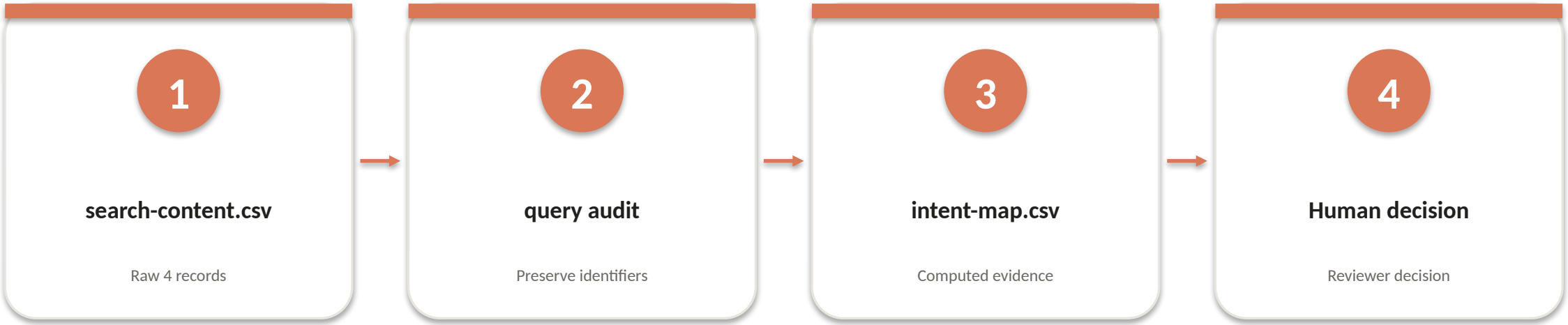
SEO Content Experiment · Control and Recovery

Control point	Risk	Required evidence
query identity	ai marketing course omitted or changed	Reconcile raw identifiers in search-content.csv
Metric contract	CTR = clicks/impressions. Training query: 48/800=6%; enquiry/click = 6/48=12.5%. High impression volume alone…	Independent calculation and named denominator
Inference check	A content plan targets the highest-impression query and ignores enquiry quality. Optimise against the…	Validate intent before production
Decision output	intent-map.csv and experiment-brief.md	Avoid keyword stuffing and unsupported claims. Record source date, search geography and device where…

WHEN IT MATTERS Avoid keyword stuffing and unsupported claims. Record source date, search geography and device where available. Seasonal and ranking changes can confound before/after comparisons.

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SEO Content Experiment · Output Handoff



OUTPUT CONTRACT

CTR = clicks/impressions. Training query: 48/800=6%; enquiry/click = 6/48=12.5%. High impression volume alone does not establish useful demand. Output: intent-map.csv and experiment-brief.md

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LO2 · ACTIVITY 11 · SYNTHETIC TEACHING DATA

SEO Content Experiment · Acceptance Evidence

✅ Expected result

CTRs 4%,3%,6%,3%; training enquiry/click 12.5%; one isolated change and explicit confounder.

IF IT DOESN'T WORK

Numbers disagree

Check header parsing,units,denominator and duplicate records.

Claim lacks a source

Add row IDs and scope; downgrade or remove unsupported inference.

Tool unavailable

Use the documented CSV fallback and state the unexecuted product boundary.

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3

LEARNING OUTCOME 3

Behaviour and Demand

Cowork evidence synthesis · Claude Code reproducible analysis

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Bounded Demand Forecast

ACTIVITY 12

Non-WSQ commercial short course · LO3

Six synthetic monthly enquiry counts are used to forecast one month. Claude Code must compare a last-value baseline and three-month moving average, preserving seasonality and short-history uncertainty.

YOU'LL BUILD

forecast.csv and forecast-assumptions.md

TOOLCHAIN

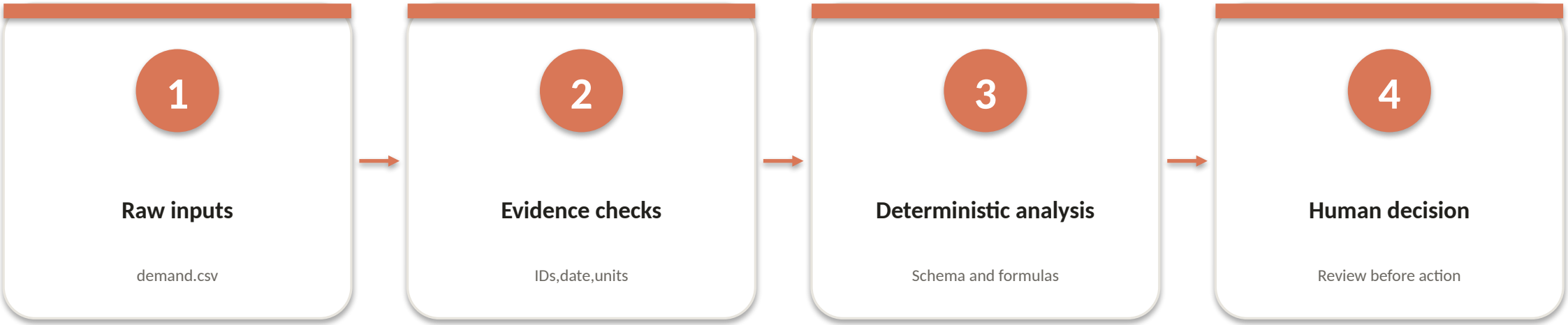
Claude Cowork · Claude Code · CSV / Markdown

DONE WHEN

Naive 104; moving-average 100; August moving-average error 10.67; campaign shift and seasonality caveats.

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Bounded Demand Forecast · Data and Trust Boundaries



CONTRACT

Check regular intervals -> reserve test period -> compute naive baseline -> compare moving average -> record error -> state scenario range -> monitor change.

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Bounded Demand Forecast · Input Schema

Field	Contract	Quality control
month	str; example: 2026-03	Preserve month;check units/identity
enquiries	int; example: 80	Preserve enquiries;check units/identity
campaign_change	str; example: no	Preserve campaign_change;check units/identity

WHEN IT MATTERS Source rows remain immutable. New computed fields belong in outputs; ambiguous data is qualified rather than invented.

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Bounded Demand Forecast · Evidence Exhibit

month	enquiries	campaign_change
2026-03	80	no
2026-04	88	no
2026-05	84	no
2026-06	96	yes
2026-07	100	no
2026-08	104	no

WHEN IT MATTERS This exhibit is an authored synthetic dataset. Read exact units and identifiers before interpreting values.

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Bounded Demand Forecast · Worked Calculation

Interpret the denominator,units and decision boundary.

```
September last-value forecast=104; three-month
mean=(96+100+104)/3=100. At August, rolling mean of
May-
July=93.33; absolute error=10.67.
```

Inputs

demand.csv

Calculation

Recompute exact values independently.

Interpretation

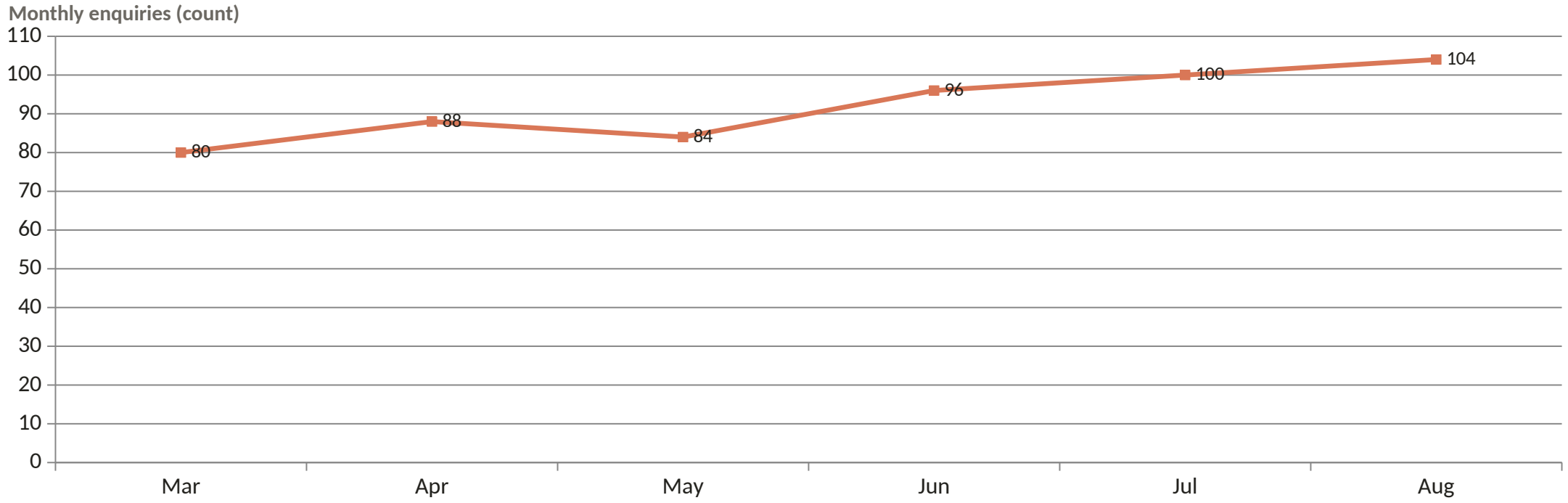
Synthetic monthly enquiries. A campaign changed in June; trend continuation is an…

Boundary

Synthetic descriptive evidence;live/causal result not established.

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Bounded Demand Forecast · Editable Chart



WHAT THE DATA SHOWS

Synthetic monthly enquiries. A campaign changed in June; trend continuation is an assumption, not evidence of stable future demand.

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Bounded Demand Forecast · Reproducible Contract

Readable excerpt; complete procedure and runnable practice remain in the guide.

```
values = [80,88,84,96,100,104]
naive = values[-1]
moving_average = sum(values[-3:])/3
aug_prediction = sum(values[2:5])/3
aug_abs_error = abs(values[5]-aug_prediction)
assert moving_average == 100
```

Data contract

demand.csv; exact field names matter.

Identity

Preserve row IDs or segment labels.

Calculation

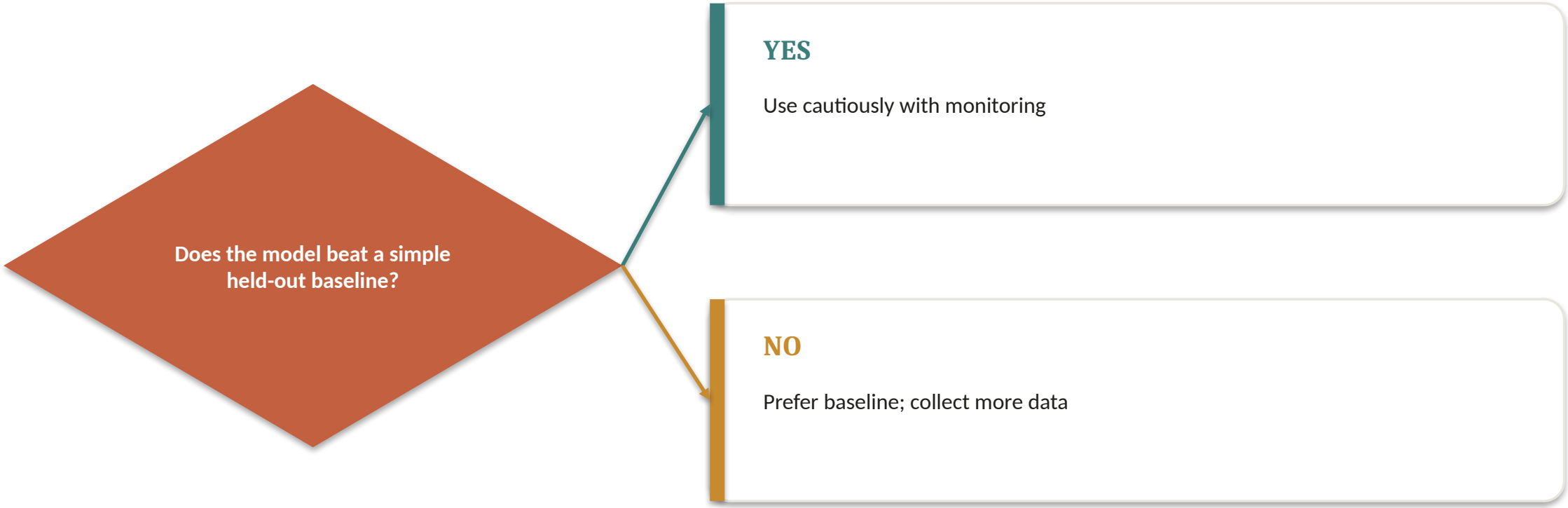
Deterministic code checks the agent narrative.

Verification

Naive 104; moving-average 100; August moving-average error 10.67; campaign shift and…

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Bounded Demand Forecast · Decision Rule



This is an authored research decision rule. It does not grant product permissions or replace human review.

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Bounded Demand Forecast · Claim-Evidence Ledger

Claim layer	Evidence required	Permitted wording
Observation	month: 2026-03	September last-value forecast=104; three-month mean=(96+100+104)/3=100. At August, rolling mean of…
Inference	Synthetic monthly enquiries. A campaign changed in June; trend continuation is an assumption, not…	Does the model beat a simple held-out baseline?
Recommendation	Use cautiously with monitoring	forecast-assumptions.md
Unknown	The agent fits an elaborate model to six points and reports a precise probability interval. The data do not…	Prefer baseline; collect more data

WHEN IT MATTERS Preserve the distinction between observed data, model output and human decision throughout Cowork and Code handoffs.

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Bounded Demand Forecast · Failure Trace

A plausible report can be wrong even when its numbers look polished.

```
The agent fits an elaborate model to six points and
reports a precise probability interval. The data do
not
support such precision.
```

Trigger

The agent promotes a claim beyond its evidence.

Diagnostic

Inspect the underlying rows,scope and denominator.

Recovery

Revise the claim and retain the rejected reasoning.

Recheck

Naive 104; moving-average 100; August moving-average error 10.67; campaign shift and…

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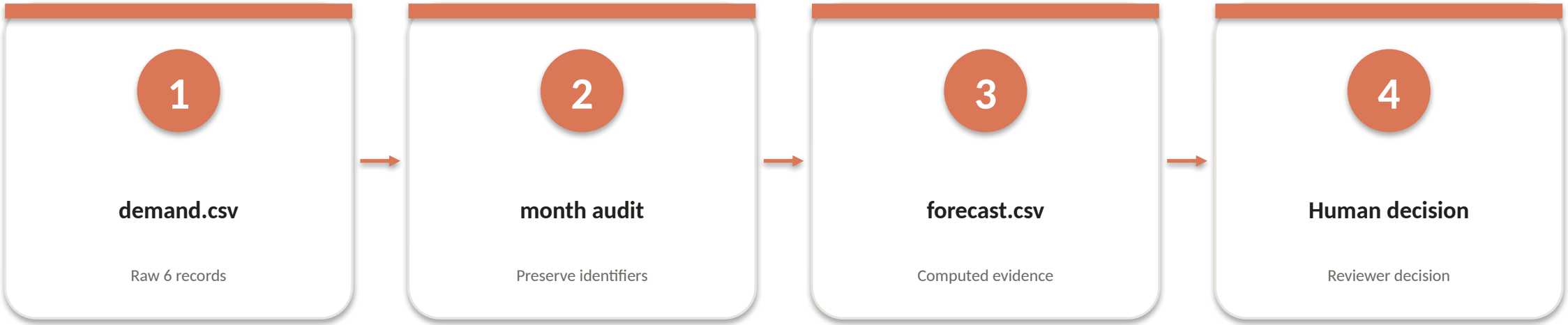
Bounded Demand Forecast · Control and Recovery

Control point	Risk	Required evidence
month identity	2026-03 omitted or changed	Reconcile raw identifiers in demand.csv
Metric contract	September last-value forecast=104; three-month mean=(96+100+104)/3=100. At August, rolling mean of…	Independent calculation and named denominator
Inference check	The agent fits an elaborate model to six points and reports a precise probability interval. The data do not…	Prefer baseline; collect more data
Decision output	forecast.csv and forecast-assumptions.md	Use simple baselines, avoid training on the held-out month, label scenario ranges as assumptions, and discuss…

WHEN IT MATTERS Use simple baselines, avoid training on the held-out month, label scenario ranges as assumptions, and discuss campaign shifts plus missing annual seasonality.

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Bounded Demand Forecast · Output Handoff



OUTPUT CONTRACT

September last-value forecast=104; three-month mean=(96+100+104)/3=100. At August, rolling mean of May-July=93.33; absolute error=10.67. Output: forecast.csv and forecast-assumptions.md

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Bounded Demand Forecast · Acceptance Evidence

✅ Expected result

Naive 104; moving-average 100; August moving-average error 10.67; campaign shift and seasonality caveats.

IF IT DOESN'T WORK

Numbers disagree

Check header parsing,units,denominator and duplicate records.

Claim lacks a source

Add row IDs and scope; downgrade or remove unsupported inference.

Tool unavailable

Use the documented CSV fallback and state the unexecuted product boundary.

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2

LEARNING OUTCOME 2

Trends and Variables

Cowork evidence synthesis · Claude Code reproducible analysis

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Claude Code Skills and MCP Audit

ACTIVITY 13

Non-WSQ commercial short course · LO2

A reproducible research workspace stores evidence, code and review rules. A local read-only MCP connector or CSV fallback supplies synthetic channel metrics; a custom Skill defines research checks without granting tools or overriding permissions.

YOU'LL BUILD

SKILL.md,analysis.py and tool-audit.md

TOOLCHAIN

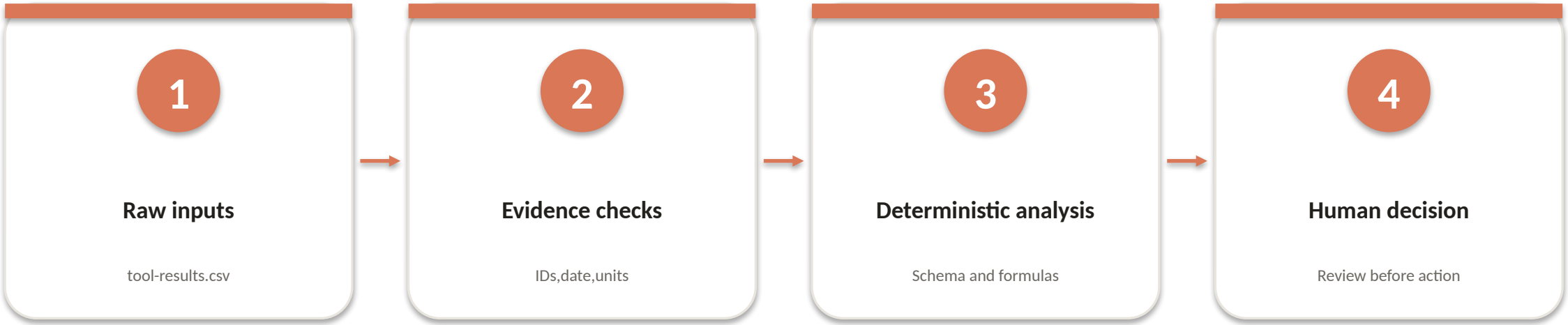
Claude Cowork · Claude Code · CSV / Markdown

DONE WHEN

Four source IDs retained; rates 5%,4%,8%,1%,overall 4%; local MCP initialize/list/call test passes; Skill front matter valid; Claude invocation separately evidenced;no live account claimed.

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Claude Code Skills and MCP Audit · Data and Trust Boundaries



CONTRACT

User question -> reviewed Skill -> approved read-only MCP tool -> schema validation -> deterministic calculation -> source-linked report -> reviewer. CSV fallback tests the same data contract.

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Claude Code Skills and MCP Audit · Input Schema

Field	Contract	Quality control
channel	str; example: Paid search	Preserve channel;check units/identity
sessions	int; example: 600	Preserve sessions;check units/identity
leads	int; example: 30	Preserve leads;check units/identity
source_id	str; example: M01	Preserve source_id;check units/identity

WHEN IT MATTERS Source rows remain immutable. New computed fields belong in outputs; ambiguous data is qualified rather than invented.

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Claude Code Skills and MCP Audit · Evidence Exhibit

channel	sessions	leads	source_id
Paid search	600	30	M01
Organic	900	36	M02
Email	300	24	M03
Direct	600	6	M04

WHEN IT MATTERS This exhibit is an authored synthetic dataset. Read exact units and identifiers before interpreting values.

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Claude Code Skills and MCP Audit · Worked Calculation

Interpret the denominator,units and decision boundary.

```
Lead rates: paid 5%,organic 4%,email 8%,direct 1%;
total
96/2400=4%. A Skill’s instructions do not create
connector access.
```

Inputs

tool-results.csv

Calculation

Recompute exact values independently.

Interpretation

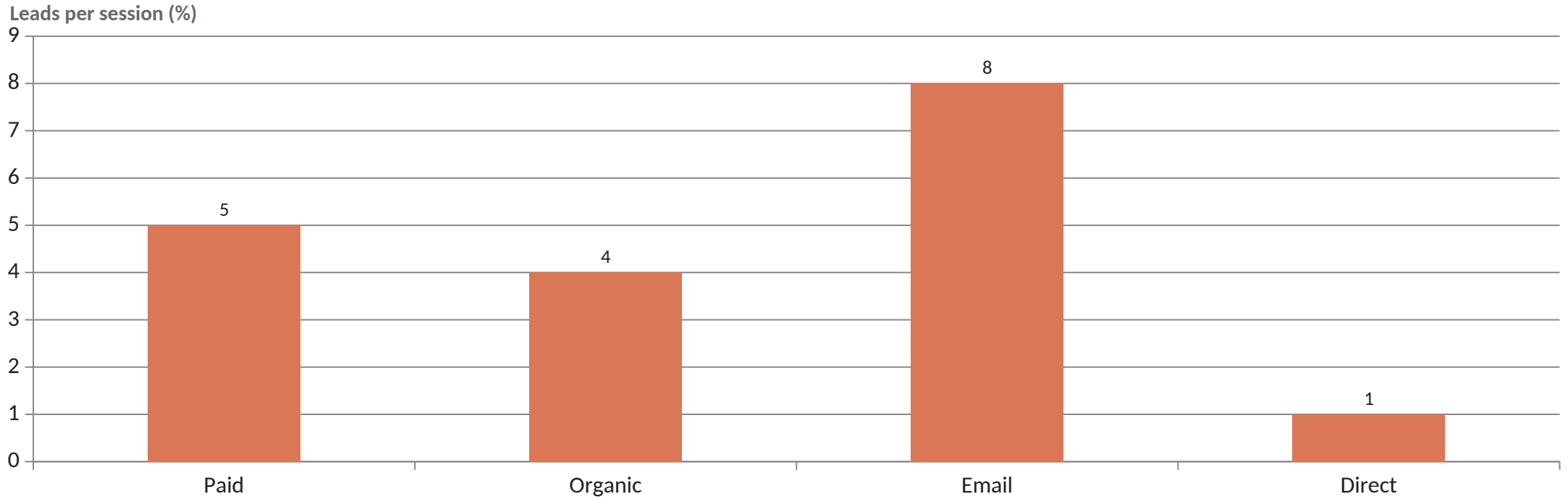
Synthetic read-only metrics. Email has a higher observed lead rate; audience…

Boundary

Synthetic descriptive evidence;live/causal result not established.

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Claude Code Skills and MCP Audit · Editable Chart



WHAT THE DATA SHOWS

Synthetic read-only metrics. Email has a higher observed lead rate; audience selection prevents a causal superiority claim.

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Claude Code Skills and MCP Audit · Reproducible Contract

Readable excerpt; complete procedure and runnable practice remain in the guide.

```
---
name: market-evidence-audit
description: Audit evidence-linked marketing research
---
Preserve source_id and raw inputs.
State denominator, window and limitations.
Reject unsupported claims; write outputs only.
```

Data contract

tool-results.csv; exact field names matter.

Identity

Preserve row IDs or segment labels.

Calculation

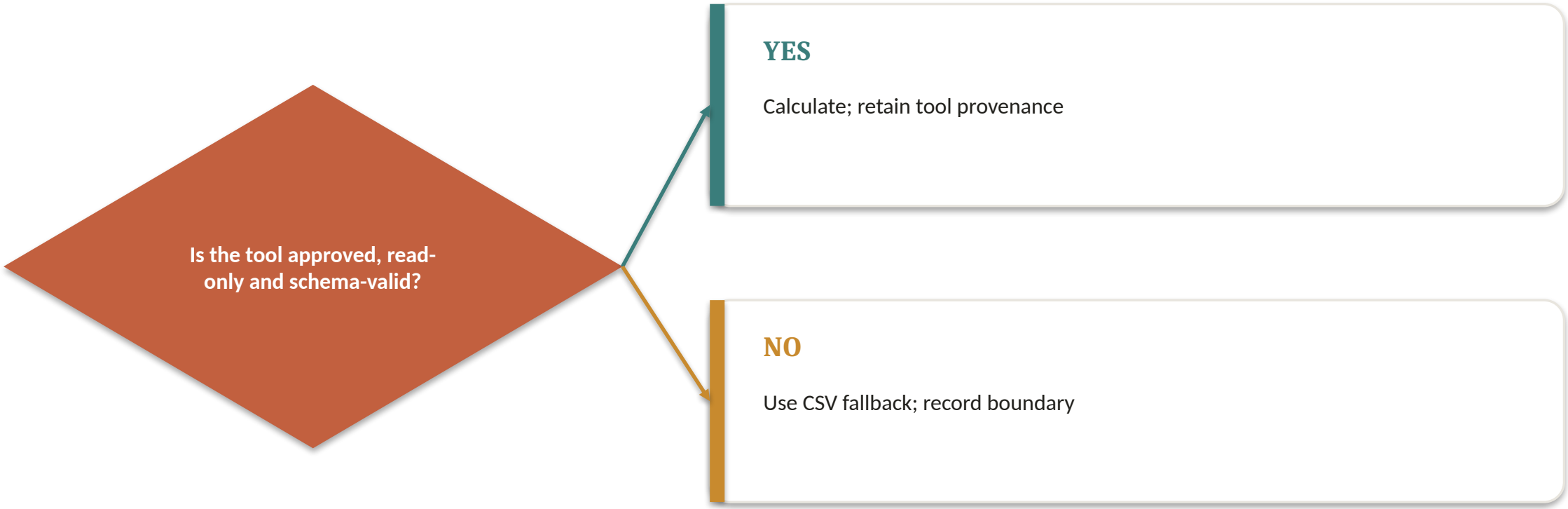
Deterministic code checks the agent narrative.

Verification

Four source IDs retained; rates 5%,4%,8%,1%,overall 4%; local MCP…

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Claude Code Skills and MCP Audit · Decision Rule



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Claude Code Skills and MCP Audit · Claim-Evidence Ledger

Claim layer	Evidence required	Permitted wording
Observation	channel: Paid search	Lead rates: paid 5%,organic 4%,email 8%,direct 1%; total 96/2400=4%. A Skill's instructions do not create…
Inference	Synthetic read-only metrics. Email has a higher observed lead rate; audience selection prevents a…	Is the tool approved, read-only and schema-valid?
Recommendation	Calculate; retain tool provenance	tool-audit.md
Unknown	The agent treats the Skill as permission to access a production account. Instruction reuse and tool…	Use CSV fallback; record boundary

WHEN IT MATTERS Preserve the distinction between observed data, model output and human decision throughout Cowork and Code handoffs.

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Claude Code Skills and MCP Audit · Failure Trace

A plausible report can be wrong even when its numbers look polished.

```
The agent treats the Skill as permission to access a
production account. Instruction reuse and tool
authorisation are separate controls.
```

Trigger

The agent promotes a claim beyond its evidence.

Diagnostic

Inspect the underlying rows,scope and denominator.

Recovery

Revise the claim and retain the rejected reasoning.

Recheck

Four source IDs retained; rates 5%,4%,8%,1%,overall 4%; local MCP…

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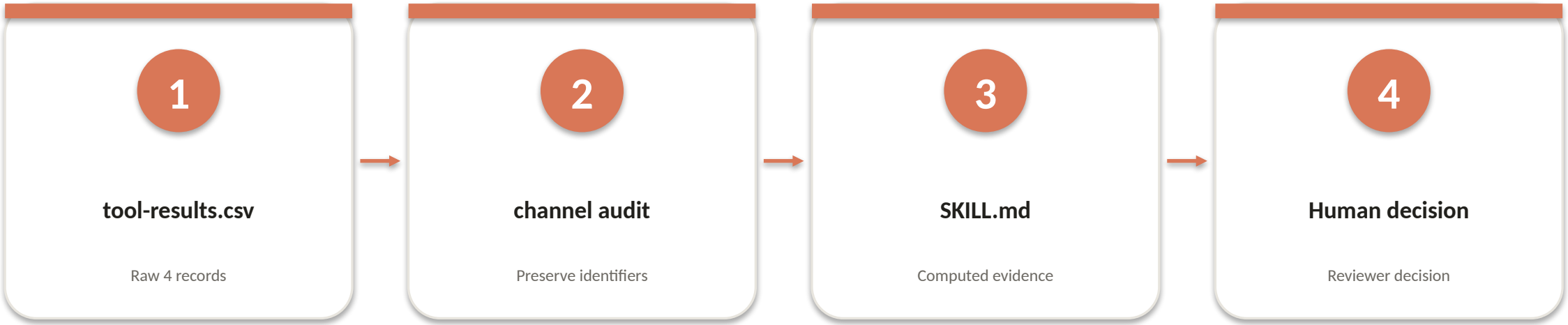
Claude Code Skills and MCP Audit · Control and Recovery

Control point	Risk	Required evidence
channel identity	Paid search omitted or changed	Reconcile raw identifiers in tool-results.csv
Metric contract	Lead rates: paid 5%,organic 4%,email 8%,direct 1%; total 96/2400=4%. A Skill's instructions do not create…	Independent calculation and named denominator
Inference check	The agent treats the Skill as permission to access a production account. Instruction reuse and tool…	Use CSV fallback; record boundary
Decision output	SKILL.md,analysis.py and tool-audit.md	Inspect tool scope and data destinations; use read-only synthetic connector, no secrets in repository…

WHEN IT MATTERS Inspect tool scope and data destinations; use read-only synthetic connector, no secrets in repository. Require approval for new network endpoints and use local fallback if account access is unavailable.

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Claude Code Skills and MCP Audit · Output Handoff



OUTPUT CONTRACT

Lead rates: paid 5%,organic 4%,email 8%,direct 1%; total 96/2400=4%. A Skill's instructions do not create connector access. Output: SKILL.md,analysis.py and tool-audit.md

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Claude Code Skills and MCP Audit · Acceptance Evidence

✅ Expected result

Four source IDs retained; rates 5%,4%,8%,1%,overall 4%; local MCP initialize/list/call test passes; Skill front matter valid; Claude invocation separately evidenced;no live account claimed.

IF IT DOESN'T WORK

Numbers disagree

Check header parsing,units,denominator and duplicate records.

Claim lacks a source

Add row IDs and scope; downgrade or remove unsupported inference.

Tool unavailable

Use the documented CSV fallback and state the unexecuted product boundary.

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4

LEARNING OUTCOME 4

Marketing Effectiveness

Cowork evidence synthesis · Claude Code reproducible analysis

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Effectiveness Decision Capstone

ACTIVITY 14

Non-WSQ commercial short course · LO4

A synthetic cross-channel campaign has spend, leads and revenue. Teams combine Cowork evidence synthesis and Claude Code calculations into a budget recommendation with attribution sensitivity, margin boundaries and a human approval record.

YOU'LL BUILD

effectiveness-report.md,budget-test.csv and approval-record.md

TOOLCHAIN

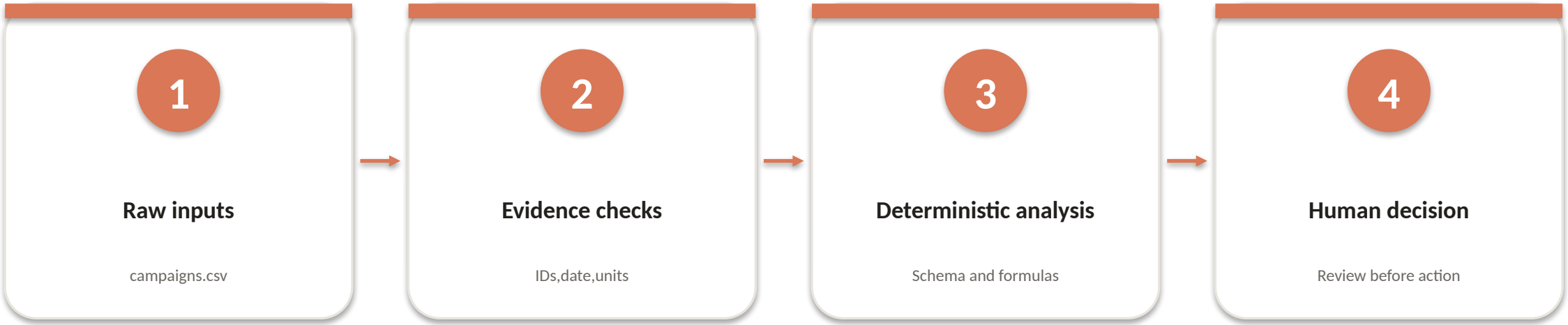
Claude Cowork · Claude Code · CSV / Markdown

DONE WHEN

96 leads; spend $2100; revenue $6000; contribution $900; paid CPL40 and contribution0; human approval pending.

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Effectiveness Decision Capstone · Data and Trust Boundaries



CONTRACT

Reconcile inputs -> compute denominator-aware metrics -> apply margin -> inspect attribution -> challenge confounders -> recommend bounded test -> human approve -> monitor.

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Effectiveness Decision Capstone · Input Schema

Field	Contract	Quality control
channel	str; example: Paid search	Preserve channel;check units/identity
spend_sgd	int; example: 1200	Preserve spend_sgd;check units/identity
leads	int; example: 30	Preserve leads;check units/identity
revenue_sgd	int; example: 2400	Preserve revenue_sgd;check units/identity
gross_margin_rate	float; example: 0.5	Preserve gross_margin_rate;check units/identity

WHEN IT MATTERS Source rows remain immutable. New computed fields belong in outputs; ambiguous data is qualified rather than invented.

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Effectiveness Decision Capstone · Evidence Exhibit

channel	spend_sgd	leads	revenue_sgd	gross_margin_rate
Paid search	1200	30	2400	0.5
Organic	600	36	1800	0.5
Email	300	24	1200	0.5
Direct	0	6	600	0.5

WHEN IT MATTERS This exhibit is an authored synthetic dataset. Read exact units and identifiers before interpreting values.

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Effectiveness Decision Capstone · Worked Calculation

Interpret the denominator,units and decision boundary.

```
Paid CPL=$1200/30=$40; revenue ROAS=2; contribution
after
spend=$2400×0.5-$1200=$0. ROAS is not profit or causal
ROI.
```

Inputs

campaigns.csv

Calculation

Recompute exact values independently.

Interpretation

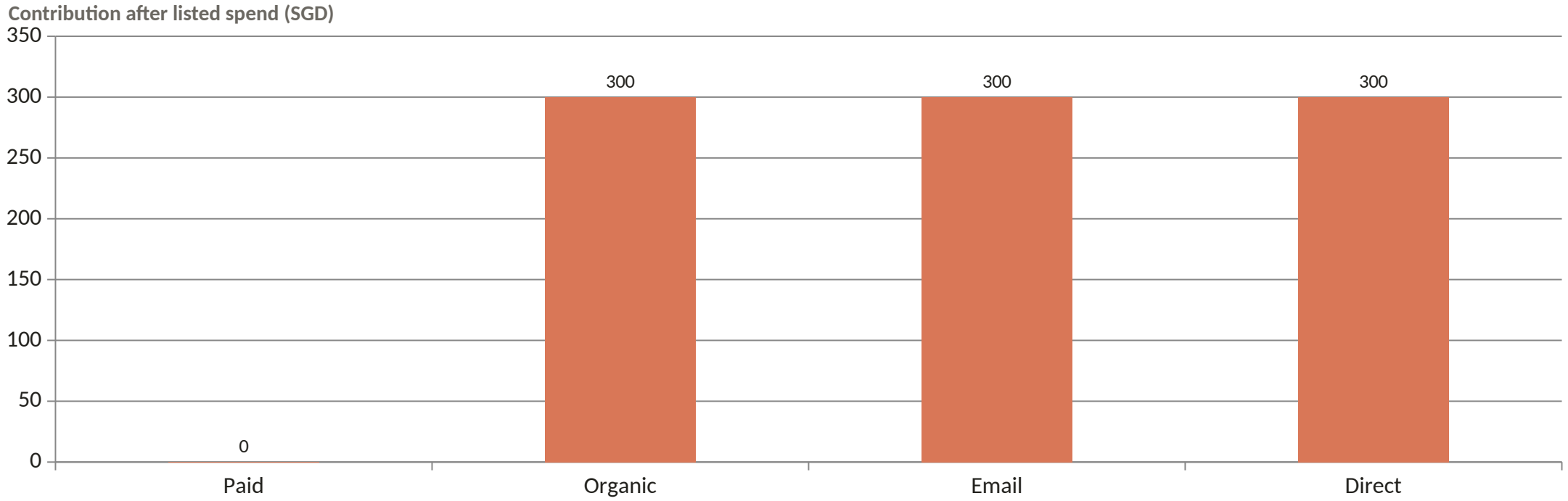
Synthetic gross-margin contribution after listed spend. Direct's zero allocated…

Boundary

Synthetic descriptive evidence;live/causal result not established.

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Effectiveness Decision Capstone · Editable Chart



WHAT THE DATA SHOWS

Synthetic gross-margin contribution after listed spend. Direct’s zero allocated spend does not mean customer acquisition is free.

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Effectiveness Decision Capstone · Reproducible Contract

Readable excerpt; complete procedure and runnable practice remain in the guide.

```
for r in rows:
    spend = float(r["spend_sgd"])
    revenue = float(r["revenue_sgd"])
    margin = float(r["gross_margin_rate"])
    r["contribution"] = revenue*margin-spend
    r["cpl"] = spend/int(r["leads"])
```

Data contract

campaigns.csv; exact field names matter.

Identity

Preserve row IDs or segment labels.

Calculation

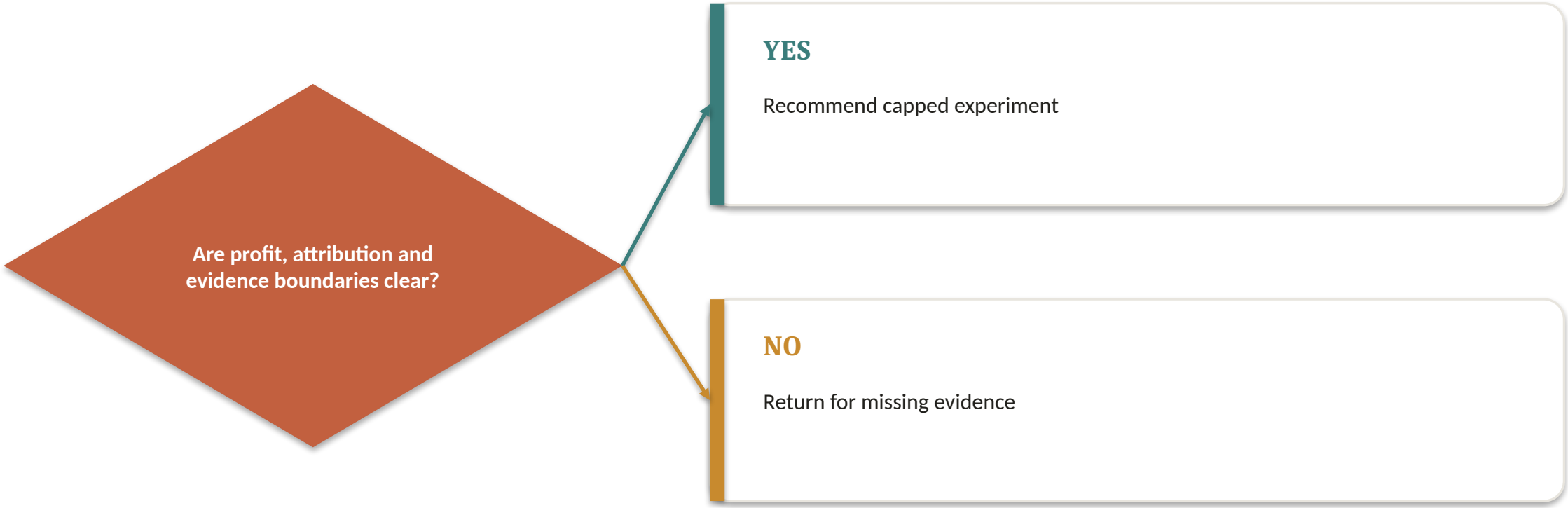
Deterministic code checks the agent narrative.

Verification

96 leads; spend $2100; revenue $6000; contribution $900; paid CPL40 and…

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Effectiveness Decision Capstone · Decision Rule



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Effectiveness Decision Capstone · Claim-Evidence Ledger

Claim layer	Evidence required	Permitted wording
Observation	channel: Paid search	Paid CPL=$1200/30=$40; revenue ROAS=2; contribution after spend=$2400×0.5-$1200=$0. ROAS is not profit or…
Inference	Synthetic gross-margin contribution after listed spend. Direct's zero allocated spend does not mean…	Are profit, attribution and evidence boundaries clear?
Recommendation	Recommend capped experiment	approval-record.md
Unknown	The report calls paid search profitable because ROAS is 2. At a 50% gross margin, contribution after media…	Return for missing evidence

WHEN IT MATTERS Preserve the distinction between observed data, model output and human decision throughout Cowork and Code handoffs.

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Effectiveness Decision Capstone · Failure Trace

A plausible report can be wrong even when its numbers look polished.

```
The report calls paid search profitable because ROAS
is
2. At a 50% gross margin, contribution after media
spend
is zero before other costs.
```

Trigger

The agent promotes a claim beyond its evidence.

Diagnostic

Inspect the underlying rows,scope and denominator.

Recovery

Revise the claim and retain the rejected reasoning.

Recheck

96 leads; spend $2100; revenue $6000; contribution $900; paid CPL40 and…

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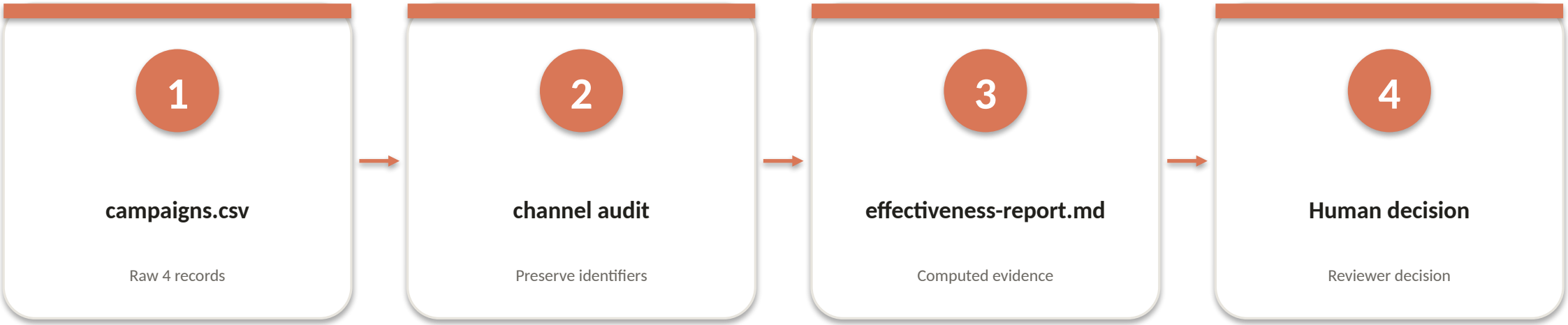
Effectiveness Decision Capstone · Control and Recovery

Control point	Risk	Required evidence
channel identity	Paid search omitted or changed	Reconcile raw identifiers in campaigns.csv
Metric contract	Paid CPL=$1200/30=$40; revenue ROAS=2; contribution after spend=$2400×0.5-$1200=$0. ROAS is not profit or…	Independent calculation and named denominator
Inference check	The report calls paid search profitable because ROAS is 2. At a 50% gross margin, contribution after media…	Return for missing evidence
Decision output	effectiveness-report.md,budget-test.csv and approval-record.md	State attribution model, accounting coverage and consent limitations. Cap recommendations to a test budget…

WHEN IT MATTERS State attribution model, accounting coverage and consent limitations. Cap recommendations to a test budget and require human sign-off before publishing or changing campaigns.

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Effectiveness Decision Capstone · Output Handoff



OUTPUT CONTRACT

Paid CPL=$1200/30=$40; revenue ROAS=2; contribution after spend=$2400×0.5-$1200=$0. ROAS is not profit or causal ROI. Output: effectiveness-report.md,budget-test.csv and approval-record.md

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Effectiveness Decision Capstone · Acceptance Evidence

✅ Expected result

96 leads; spend $2100; revenue $6000; contribution $900; paid CPL40 and contribution0; human approval pending.

IF IT DOESN'T WORK

Numbers disagree

Check header parsing,units,denominator and duplicate records.

Claim lacks a source

Add row IDs and scope; downgrade or remove unsupported inference.

Tool unavailable

Use the documented CSV fallback and state the unexecuted product boundary.

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Behavioural Analytics and Primary Research

Evidence type	What it can support	Missing evidence
GA4 consented events	Observed digital actions and trends	Motivation,non-visitors,causality
Survey/interview	Stated needs and willingness	Observed behaviour and selection bias
Cowork synthesis	Evidence organisation and drafts	Truth without source verification
Claude Code	Reproducible checks and models	Valid assumptions and data rights

WHEN IT MATTERS K1 · Combine behavioural traces with primary research; agent output is an analysis artifact, not a new source.

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Demographic Segmentation and Bias

Variable	Permitted use	Limit
Self-reported role	Tailor validation questions	Missing and volunteer bias
Age band with consent	Aggregate audience research	Avoid identifying small groups
Observed journey stage	Behavioural messaging hypothesis	Intent remains inferred
Unknown attribute	Leave unknown	Never infer protected traits

WHEN IT MATTERS K2 · Demographic and behavioural variables answer different questions; do not invent persona attributes.

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Pricing Evidence and Willingness to Pay

Evidence	Calculation or mechanism	Boundary
Competitor offer	SGD/hour with matched scope	Advertised price is not willingness to pay
Checkout behaviour	Completed purchases / eligible users	Confounded by acquisition and seasonality
Survey question	Neutral stated-price response	Hypothetical and selection bias
Controlled price test	Randomised eligible groups	Ethics,consent and business approval

WHEN IT MATTERS K3 · GA4 can describe behaviour around offers; independent pricing research and controlled tests are needed for demand interpretation.

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FILED K1–K8 / A1–A8 COVERAGE · SYNTHETIC EXAMPLES

Survey Questions That Test a Decision

Question design	Weak item	Improved item
Timing	Would you love evening classes?	Which session time can you attend?
Content need	Is AI marketing amazing?	Which task is hardest this month?
Pricing	Would you pay our great price?	Which option would you consider at $X?
Bias check	Existing learners only	Include target non-customers and record sampling

WHEN IT MATTERS K4/A3 · Define decision,population,response options and sampling limits before agent-assisted coding.

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GA4 Account-to-Stream Boundary

Object	Responsibility	Research implication
Account	Organisational ownership/access	Confirm authorised property
Property	Web/app reporting boundary	Do not mix unrelated brands
Data stream	Website/app collection input	Confirm stream and consent scope
Export	Field/window selection	Record source,date,filters and units

WHEN IT MATTERS Legacy setup coverage retained as architecture; exact click paths are in the LG and current official documentation.

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Dimensions, Metrics and Denominators

Contract	Example	Control
Dimension	session source / medium	Choose matching attribution scope
Metric	sessions;event count	Do not call event count users
Rate	key events / stated denominator	State user/session/event denominator
Filter	period,channel,device	Preserve filter state with export

WHEN IT MATTERS LO1/LO2 · A report export is interpretable only when its dimension,metric,scope and filter contract is retained.

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GA4 Acquisition and Trend Contract

Analysis	Input	Evidence boundary
User acquisition	First user source/medium	Initial acquisition perspective
Traffic acquisition	Session source/medium	Session perspective
Trend	Same metric and intervals	Account for tracking or campaign changes
Competitor keyword	External licensed/public source	Not a GA4 competitor feed

WHEN IT MATTERS K5/A4 · Compare aligned periods and source scope; obtain competitor evidence separately.

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GA4 Engagement and Behaviour Indicators

Indicator	Interpretation	Limit
Engaged sessions	Qualified session behaviour	Definition and consent coverage matter
Views	Page/screen event volume	Not unique audience
Key events	Marked important outcomes	Trigger quality must be tested
Average engagement	Engagement metric within scope	Does not establish motivation

WHEN IT MATTERS A5/A6 · Report exact metric labels and period; benchmark only comparable populations and methods.

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GA4 Explorations: Variables to Technique

Technique	Contract	Decision
Free form	Dimensions,metrics,segments	Inspect cross-tabs and distributions
Funnel	Ordered steps,open/closed,window	Locate observed drop-off
Path	Start/end node and event/page type	Explore sequence patterns
External forecast	Exported regular time series	Not claimed GA4-native forecasting

WHEN IT MATTERS K7 · Explorations describe observed data; external models must keep assumptions and execution provenance.

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Forecast Model Comparison

Model	Held-out August prediction	Absolute error
Last value	July=100	|104-100|=4
Three-month mean	(84+96+100)/3=93.33	10.67
Complex model	Six observations insufficient	Unsupported precision
September choice	Naive104 or MA100	Baseline comparison favours naive here

WHEN IT MATTERS K6/A7 · Compare predictions on a held-out period,then forecast; six months cannot resolve annual seasonality.

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Descriptive, Predictive and Causal Models

Technique	Question	Evidence
Descriptive funnel/segment	Where/how do groups differ?	Aligned counts and definitions
Time-series baseline	What next-period demand?	Held-out error and drift assumptions
Controlled experiment	Did a change cause lift?	Random assignment and guardrails
Attribution allocation	Who receives credit?	Model and lookback sensitivity

WHEN IT MATTERS K7 · Distinguish descriptive,predictive,causal and allocation models before evaluating marketing effectiveness.

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Controlled Creative Experiment

Contract	Synthetic example	Failure control
Population	400 eligible users per arm	Keep eligibility equal
Assignment	Random A/B before exposure	Avoid self-selected arms
Outcome	A20/400=5%;B28/400=7%	+2 percentage points observed
Boundary	Small sample;uncertain lift	Do not claim guaranteed improvement

WHEN IT MATTERS A8 · Specify unit,control,primary metric,window and stop rule; significance needs a suitable statistical analysis.

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Conversion Rate, CPA and Revenue ROAS

Metric	Worked example	Boundary
Conversion rate	12 enrolments/2400 sessions=0.5%	State cohort and denominator
CPA	Spend1200/12 acquisitions=100	Acquisitions differ from leads
CPL	Spend1200/30 leads=40	Lead is not a paying customer
Revenue ROAS	Attributed revenue2400/spend1200=2	Not profit or incremental ROI

WHEN IT MATTERS K8 · Contribution after spend=2400×50%-1200=0; protect metric names and accounting boundaries.

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GA4 Key Events and Ads Conversions

State	Meaning	Control
Collected event	Received interaction	Validate trigger and parameters
Key event	Important business outcome in GA4	Mark according to measurement plan
Ads conversion	Advertising measurement outcome	Product linkage and configuration matter
Report evidence	Current property,period,metric	Do not reuse legacy conversion labels blindly

WHEN IT MATTERS LO4 · Current GA4 distinguishes key events from Ads conversions; a drafted contract is not live configuration evidence.

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FILED K1–K8 / A1–A8 COVERAGE · SYNTHETIC EXAMPLES

Import and Join Integrity

Join contract	Synthetic example	Risk
Cost key	date+source+medium+campaign	Unmatched cost rows distort ROAS
Item key	item_id	Missing catalog match
User data	Approved pseudonymous identifier	Consent and re-identification risk
Reconciliation	Matched + unmatched = raw rows	Keep rejected records visible

WHEN IT MATTERS Legacy data-import coverage retained as a data contract; no real customer identifiers are used in training.

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RESEARCH METHOD · ILLUSTRATIVE GENERATED IMAGE

Primary Evidence:Question,Response and Coding



Decision:evening programme demand

Neutral question:Which session times can you attend?

Response record:participant_id,time preference,quote,consent status.

Code:timing;preserve exact quote and non-response count.

Generated illustrative photograph;no real participant or collected evidence. Synthetic personas are hypotheses,not replacements for representative respondents.

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COURSE CLOSING

Course Feedback

1. **Feedback** — Share your views
2. **Reflection** — What you will apply
3. **Questions** — Open discussion
4. **Resources** — Slides and activities
5. **Next steps** — Continue practising

THANK YOU

Tell us what worked and what to improve, so the next run is better.

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TERTIARY INFOTECH ACADEMY

Thank You

Keep the evidence trail:source -> calculation -> limitation -> reviewed decision.

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